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Cayley incidence graphs*

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Abstract

Evra, Feigon, Maurischat, and Parzanchevski (2023) introduced a biregular extension of Cayley graphs. In this paper, we reformulate their definition and provide some basic properties. We also show how these Cayley incidence graphs relate to various notions of Cayley hypergraphs. We further establish connections between Cayley incidence graphs and certain geometric and combinatorial structures, including coset geometries, difference sets, and cages.

Keywords: Cayley graphs, bipartite graphs, hypergraphs, designs, automorphisms.

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1 Introduction

Cayley graphs were introduced by Arthur Cayley [10] in the late 19th century with the objective of illustrating abstract groups. They have now become a widely studied topic in both group theory and graph theory as they connect these two areas of mathematics in a beautiful way. A Cayley graph is defined by a group and a subset of this group. The group is contained in the automorphism group of the graph and acts regularly on the vertices. In particular, Cayley graphs are vertex-transitive, which makes them a well-behaved class of graphs to work with.

A key application of Cayley graphs is to explicitly construct graph expanders. Expanders are infinite families of sparse graphs with high connectivity that have a number of applications [35]. They can equivalently be seen as families of graphs where the absolute values of the non-trivial eigenvalues are small.

Alon and Boppana [2] and Serre [51] proved that the limit of the sequence of second-largest eigenvalues of an infinite family of k-regular graphs is at least $2\sqrt{k-1}$. Graphs that meet this bound are called *Ramanujan graphs*, and Lubotzky, Philips, and Sarnak [42], Margulis [45], and Morgenstern [47], among others, used Cayley graphs to explicitly construct k-regular Ramanujan graphs for certain values of k.

For an infinite family of bipartite biregular graphs, the corresponding bound comes from Feng and Li [23]. Marcus, Spielman, and Srivastava [44] proved for any k and ℓ, there exists an infinite family of (ℓ, k)-biregular Ramanujan graphs. Their proof is probablistic, not constructive, and Evra, Feigon, Maurischat, and Parzanchevski [21] introduced what they called Cayley bigraphs to explicitly construct the first bipartite biregular Ramanujan graphs. These Cayley bigraphs are the topic of this paper, but to avoid confusion with Cayley digraphs, we will refer to them as Cayley incidence graphs.

Proving that an infinite sequence of graphs is a family of expanders generally requires deep tools from representation theory and number theory. However, Cayley graphs can also be used in a more sporadic way to construct examples of graphs with certain structural and spectral properties. In Proposition 3.3, we show that Cayley incidence graphs have similar potential by showing that we can compute the spectrum of a Cayley incidence graph at least as easily as we can compute the spectrum of a Cayley graph. Although this is not trivial, computing the spectra of Cayley graphs has been well studied. In 1979, Babai [4] derived an expression for the eigenvalues of a Cayley graph in terms of the irreducible characters of the group. More has been done since then and a survey on the topic was published recently [41].

Any bipartite graph can be interpreted as the incidence graph of a hypergraph, and several different hypergraph generalizations of Cayley graphs have been introduced over the years [9, 38, 40, 52, 55]. A particular axiom defined by Evra et al [21] imposes additional structure on the incidence graph, and in Corollary 4.5, we show that Cayley incidence graphs are precisely the restriction of the hypergraphs of Shee [52], Lee and Kwon [40], or Jajcay and Jajcayová [38] to linear uniform regular hypergraphs.

Cayley incidence graphs also relate to structures of interest in design theory and finite geometry. In Proposition 5.2 we show that $(n, k, 1)$-difference sets [39, 54] correspond to a certain subclass of Cayley incidence graphs. In particular, it implies that Desarguesian finite projective planes can be represented as Cayley incidence graphs. We show this explicitly by giving constructions of finite projective and affine spaces of any dimension over any finite field as Cayley incidence graphs in Example 5.3 and Example 5.4.

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Another example comes from the coset geometries introduced by Tits [56] to characterize groups. In Proposition 7.1 and Example 7.3, we illustrate how certain kinds of Cayley incidence graphs relate to a particular class of coset geometries.

If we restrict ourselves to Cayley graphs which are bipartite, a natural question is whether they are related to Cayley incidence graphs. Indeed, as we show in Theorem 9.2, a bipartite Cayley graph of girth at least six can be represented as a Cayley incidence graph. However, there are some Cayley incidence graphs which cannot be constructed as bipartite Cayley graphs. Related to Cayley graphs are bi-Cayley graphs, which admit a semiregular group of automorphisms with two orbits. These graphs have been widely researched and they are significant in studying the symmetry of graphs [15, 16]. In Theorem 9.8 we show that a bipartite bi-Cayley graph with girth at least six is isomorphic to a Cayley incidence graph.

Cayley incidence graphs are a new class of graphs with close connections to generalizations of Cayley graphs studied in various contexts. We conclude the paper by highlighting some open problems of particular interest, as much of the theory and applications of these graphs remains open for further study.

2 Definition

Throughout this paper, we will generally assume that G is a multiplicative group with identity e. However, when dealing with specific groups it will often be convenient to use the notation for that group as, for instance, using additive notation and identity 0 when dealing with cyclic groups.

For a set C of elements of G, we will let

$$gC := \{gs : s \in C\}.$$

Let S be an inverse-closed subset of $G \setminus \{e\}$. The *Cayley graph*, $\mathrm{Cay}(G, S)$, is the graph on vertex set G with vertices g, h adjacent if and only if $h^{-1}g \in S$. The set S is called the *connection set* of $\mathrm{Cay}(G, S)$.

Let G be a group, and let $\pi = \{C_1, \ldots, C_\ell\}$ be a collection of subsets of G that each contains the identity. For future notational convenience, we let $\gamma := G$ and define

$$\beta := \{gC : g \in G, C \in \pi\}.$$

We wish to define an incidence relation between γ and β. If $g \in \gamma$, we can define an incidence relation $g \sim_1 gC_i$ for $1 \leq i \leq \ell$. If $gC \in \beta$, we can define an incidence relation $gC \sim_2 gs$ for all $s \in C$. Both these incidence relations are natural, but they are not generally equivalent.

Example 2.1. Let $G = \mathbb{Z}_5$ and let $\pi = \{\{0, 1, 2\}, \{0, 3, 4\}\}$. Using $\sim_1$, we see that 1 is incident to $\{1, 2, 3\}$ and $\{0, 1, 4\}$. However, using $\sim_2$, we have that 1 is incident to $\{0, 1, 2\}$, $\{1, 2, 3\}$ and $\{0, 1, 4\}$.

Fix $D \in \beta$. It is incident to the same set of elements from G due to $\sim_1$ and $\sim_2$ precisely when, for every $h \in D$, there exists $1 \leq j \leq \ell$ such that $h^{-1}D = C_j$. We can reframe this as a condition on π.

Definition 2.2 (T-axiom)**.** Let G be a group and let π be a collection of subsets of G that each contain the identity. We say it satisfies the *T-axiom* if for any $C \in \pi$ and $s \in C$, we have $s^{-1}C \in \pi$.

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The T in T-axiom stands for "translate" as it imposes a condition on translates of $C \in \pi$. We will refer to the sets $C \in \pi$ as *cells*. We can now use our singular incidence relation to define a bipartite incidence graph between group elements and translates of the cells for a particular choice of π.

Definition 2.3 (Cayley incidence graph)**.** Let G be a group, let $\pi = \{C_1, \ldots, C_\ell\}$ be a collection of subsets of G of size k that intersect pairwise precisely in the identity and suppose π satisfies the T-axiom. Let

$$\gamma = G \text{ and } \beta = \{gC : g \in G, C \in \pi\}.$$

We define the *Cayley incidence graph* $\mathrm{Cin}(G, \pi)$ as the bipartite graph on vertex set $\gamma \cup \beta$ with an edge from g to gC for all $g \in G$ and $C \in \pi$.

Note that $\mathrm{Cin}(G, \pi)$ is biregular. An (ℓ, k)-biregular Cayley incidence graph has ℓ cells of size k.

Example 2.4. If $\ell = 1$, then π has a single cell C. This satisfies the T-axiom precisely when C is a subgroup. In this case $\mathrm{Cin}(G, \{C\})$ is a disjoint union of $K_{1,k}$.

Example 2.5. More generally, whenever π is a collection of ℓ subgroups of G of order k that intersect pairwise in the identity, the T-axiom is trivially satisfied, so $\mathrm{Cin}(G, \pi)$ is well-defined.

For any π coming from a Cayley incidence graph, let

$$S(\pi) = \bigcup_{i=1}^{\ell} C_i \setminus \{e\}.$$

It is easy to see that $S(\pi)$ is an inverse-closed subset of $G \setminus \{e\}$, so we may define the Cayley graph $\mathrm{Cay}(G, S(\pi))$. This is the *underlying Cayley graph* of $\mathrm{Cin}(G, \pi)$.

Example 2.6. If $k = 2$, then $\pi = \{\{e, g_1\}, \{e, g_2\}, \ldots, \{e, g_\ell\}\}$. Then π satisfies the T-axiom precisely when $S(\pi)$ is inverse-closed. In this case, $\mathrm{Cin}(G, \pi)$ is the subdivision graph of the underlying Cayley graph $\mathrm{Cay}(G, S(\pi))$.

Note that any $gC \in \beta$ corresponds to a clique in the underlying Cayley graph: if $gh_1, gh_2 \in gC$, then by the T-axiom $h_2^{-1}C \in \pi$, so

$$h_2^{-1}g^{-1}gh_1 = h_2^{-1}h_1 \in h_2^{-1}C \subseteq S(\pi).$$

In particular, π corresponds to a set of ℓ cliques of size k in $\mathrm{Cay}(G, S(\pi))$ pairwise intersecting in the identity.

The cases where $\ell = 1$ or $k = 2$ are trivial, and we will generally be interested in the non-trivial examples with $\ell \geq 2$ and $k \geq 3$.

Example 2.7 (Fano Plane)**.** Let $G = \mathbb{Z}_7$ and $\pi = \{\{0, 1, 3\}, \{0, 2, 6\}, \{0, 4, 5\}\}$. We can verify that this satisfies the T-axiom. The resulting graph, shown in Figure 1, is the smallest non-trivial Cayley incidence graph. Sometimes called the *Heawood graph*, it can also be viewed as the incidence graph of the Fano plane.

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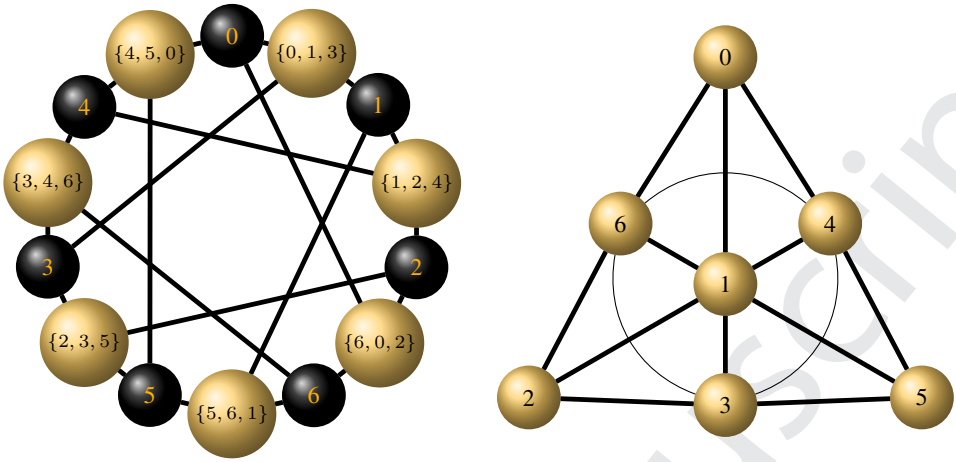


Figure 1: $\mathrm{Cin}(\mathbb{Z}_7, \{0,1,3\}, \{0,2,6\}, \{0,4,5\})$ (Heawood graph) and the Fano plane

Analogously to Cayley graphs, a Cayley incidence graph $\mathrm{Cin}(G, \pi)$ is connected if and only if $S(\pi)$ generates G. If $S(\pi)$ does not generate G, then we can without loss of generality let K be the group generated by $S(\pi)$ and consider the connected graph $\mathrm{Cin}(K, \pi)$, which is isomorphic to each connected component of $\mathrm{Cin}(G, \pi)$. Thus, when we refer to non-trivial Cayley incidence graphs, we assume that they are connected.

Let G be a group, and let $\pi = \{C_1, \ldots, C_\ell\}$ be a collection of subsets of G such that for $1 \leq i \leq \ell$ the cell C_i has size k, contains the identity, and any two distinct cells intersect in precisely the identity. Suppose further that π satisfies the T-axiom and $k \leq \ell$. We can define a partition $\hat{\pi}$ of $S(\pi) \subseteq G$ by removing the identity from every cell in π. Such a partition satisfies:

(1) The set $S(\pi)$ is inverse-closed and does not contain the identity and

(2) Whenever s and t are in the same cell $C \in \hat{\pi}$, the product $s^{-1}t$ is in the cell containing s^{-1}.

Conversely, given a group G and a partition $\hat{\pi}$ of some $S \subseteq G$ into ℓ cells of size $k-1$ satisfying (1) and (2), we can add the identity to every cell $C \in \hat{\pi}$ to get collection π satisfying the T-axiom. Evra, Feigon, Maurischat, and Parzanchevski [21] defined the bi-Cayley axioms as (1) and (2), but we find it more convenient to work with the collection π where every cell contains the identity and use the equivalent formulation of Definition 2.2. We call this the T-axiom to avoid confusion with the bi-Cayley graphs discussed in Section 9.

Evra, Feigon, Maurischat, and Parzanchevski defined an equivalence relation on $G \times \hat{\pi}$ and used this to define a bipartite graph on $\gamma \cup \beta$ with $\gamma = G$ and β the equivalence classes of their relation. They referred to the resulting graphs as *Cayley bigraphs*, and they are precisely the Cayley incidence graphs with the additional assumption that $k \leq \ell$.

Another perspective is to view a Cayley incidence graph $\mathrm{Cin}(G, \pi)$ as the incidence graph of a hypergraph. The corresponding hypergraph $\mathrm{CH}(G, \pi)$ is a k-uniform, ℓ-regular hypergraph on vertex set $\gamma = G$ with hyperedge set

$$\beta = \{gC : g \in G, C \in \pi\}.$$

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Any graph can be viewed as a 2-uniform hypergraph. As we saw in Example 2.6, a Cayley graph $\mathrm{Cay}(G, S)$ is precisely the 2-uniform hypergraph $\mathrm{CH}(G, \{\{e, s\} : s \in S\})$. Thus viewing a Cayley incidence graph as the incidence graph of a hypergraph gives a hypergraph generalization of Cayley graphs.

This is far from the first hypergraph generalization of Cayley graphs [9, 38, 40, 52], and we will explore the varying definitions and their relationship to Cayley incidence graphs in Section 4. First, it will be convenient to set up more properties of Cayley incidence graphs.

3 Basic properties

Let $X = (\gamma \cup \beta, E)$ be any bipartite graph. The *biadjacency matrix* N of X is the $\gamma \times \beta$ matrix with (u, v) entry equal to 1 if u is incident to v and 0 otherwise. The adjacency matrix has the form

$$\begin{pmatrix} \mathbf{0} & N \\ N^T & \mathbf{0} \end{pmatrix}.$$

Consider the distance-two graph X_2 of X. This is the graph on the vertex set of X with two vertices adjacent in X_2 if and only if they are at distance two in X. Since X is bipartite, the distance-two graph can be written as a disjoint union of a graph on vertex set γ and a graph on vertex set β. These graphs are the *halved graphs*, denoted H_γ and H_β.

In general, the adjacency matrix of H_γ is given by the matrix with 1 on every nonzero off-diagonal entry of NN^T and 0 everywhere else. The next proposition tells us that for a Cayley incidence graph, the halved graph H_γ is the underlying Cayley graph, with a particularly strong relationship between the adjacency matrix of H_γ and NN^T.

Proposition 3.1. *Let* $\mathrm{Cin}(G, \pi)$ *be an* (ℓ, k)*-biregular Cayley incidence graph with biadjacency matrix* N *and let* A *be the adjacency matrix of the underlying Cayley graph* $\mathrm{Cay}(G, S(\pi))$*. Then*

$$NN^T = A + \ell I.$$

Proof. Observe that NN^T counts the number of walks of length two between elements of G. Thus, if $g, h \in G$, we have that

$$\left(NN^T\right)_{g,h} = |\{gC_i : h \in gC_i\}| = \left|\left\{C_i : g^{-1}h \in C_i\right\}\right| = \begin{cases} \ell & \text{if } h = g, \\ 1 & \text{if } g^{-1}h \in S(\pi), \\ 0 & \text{if } g^{-1}h \notin S(\pi). \end{cases}$$

Recalling the definition of Cayley graphs and translating back into matrices gives the desired result. □

We defined the underlying Cayley graph in terms of the group and the connection set. Proposition 3.1 gives another way of interpreting the relationship between the Cayley incidence graph and its underlying Cayley graph, which is particularly useful in computing the spectrum.

We will need the following fact from linear algebra, which can be found, for instance, in Section 10.3 of [32].

Lemma 3.2. *Let* N *be a matrix. Then the nonzero eigenvalues of* $N^T N$ *and* NN^T *are the same, with the same multiplicity.*

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Proposition 3.3. *The spectrum of a Cayley incidence graph is determined by the spectrum of the underlying Cayley graph, the order of the group, and the valencies k and ℓ.*

Proof. Let A be the adjacency matrix of a Cayley incidence graph $\mathrm{Cin}(G,\pi)$ and let N be the biadjacency matrix. We have

$$A^2 = \begin{pmatrix} NN^T & \mathbf{0} \\ \mathbf{0} & N^TN \end{pmatrix}.$$

Let the spectrum of the underlying Cayley graph $\mathrm{Cay}(G,S(\pi))$ be

$$\left\{\theta_1^{(m_1)},\ldots,\theta_{t-1}^{(m_{t-1})},\theta_t^{(m_t)}\right\}$$

where θ_r is an eigenvalue and m_r is the multiplicity of θ_r for all $1 \leq r \leq t$. By Proposition 3.1, we see that

$$\left\{(\theta_1+\ell)^{(m_1)},(\theta_2+\ell)^{(m_2)},\ldots,(\theta_t+\ell)^{(m_t)}\right\}$$

are the eigenvalues of NN^T.

By Lemma 3.2, we know that NN^T and N^TN share nonzero eigenvalues with multiplicity. In particular, if $\theta_r+\ell$ is a nonzero eigenvalue of NN^T with multiplicity m_r, then it is an eigenvalue of A^2 with multiplicity $2m_r$. All that remains is to determine the multiplicity of zero as an eigenvalue for A^2.

If n is the order of G, then γ has size n and β has size $b = n\ell/k$. We suppose for notational convenience that $\theta_t = -\ell$, possibly with multiplicity $m_t = 0$. The multiplicity of 0 in N^TN is $m_t + b - n$, so the multiplicity of 0 as an eigenvalue in A^2 is $2m_r + b - n$.

Finally, the spectrum of a bipartite graph is symmetric over the real axis, so we get that the spectrum of $\mathrm{Cin}(G,\pi)$ is

$$\left\{\pm\sqrt{\theta_1+\ell}^{(m_1)},\ldots,\pm\sqrt{\theta_{t-1}+\ell}^{(m_{t-1})},0^{(2m_t+b-n)}\right\}. \qquad \square$$

The *girth* of a graph is the length of a shortest cycle in it. Proposition 3.1 gives us information about the girth of a Cayley incidence graph.

Lemma 3.4. *A Cayley incidence graph has girth at least six.*

Proof. Let $u,v \in G$, and recall that $\left(NN^T\right)_{u,v}$ counts the number of walks of length two from u to v. Then for $u \neq v$, we see from Proposition 3.1 that the number of paths from u to v is at most one. In particular, there cannot be a cycle of length four that goes through u, and since this is independent of our choice of u, there cannot be a cycle of length four in the graph. $\square$

Lemma 3.4 implies that every Cayley incidence graph is the incidence graph of a combinatorial configuration, as defined in [33].

Given two Cayley graphs, $\mathrm{Cay}(G_1,S_1)$ and $\mathrm{Cay}(G_2,S_2)$, their Cartesian product is a Cayley graph for the group $G_1 \times G_2$ with connection set $(S_1 \times \{e\}) \cup (\{e\} \times S_2)$. Thus, there is a natural way of constructing a Cayley graph given two smaller ones. Similarly, we can construct Cayley incidence graphs from smaller Cayley incidence graphs.

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Lemma 3.5. *Suppose that* $\mathrm{Cin}(G_1, \pi_1)$ *is an* (ℓ_1, k)*-biregular Cayley incidence graph and* $\mathrm{Cin}(G_2, \pi_2)$ *is an* (ℓ_2, k)*-biregular Cayley incidence graph. We define*

$$\pi := \{C \times \{e\} \colon C \in \pi_1\} \cup \{\{e\} \times C \colon C \in \pi_2\}.$$

Then $(G_1 \times G_2, \pi)$ *satisfies the* T*-axiom and* $\mathrm{Cin}(G_1 \times G_2, \pi)$ *has edges of one of the following forms:*

(1) $\{(g_1, h), g_2C_i \times \{h\}\}$ *where* $\{g_1, g_2C_i\}$ *is an edge of* $\mathrm{Cin}(G_1, \pi_1)$*;*

(2) $\{(h, g_1), \{h\} \times g_2C_i]\}$ *where* $\{g_1, g_2C_i\}$ *is an edge of* $\mathrm{Cin}(G_2, \pi_2)$.

Proof. It is clear by construction that π satisfies the T-axiom. Now, suppose that in $\mathrm{Cin}(G_1 \times G_2, \pi)$ we have an edge between (g_1, h_1) and $g_2C_i \times \{h_2\}$. This statement is equivalent to $h_1 = h_2$ and $g_1 \in g_2C_i$, which is equivalent to the fact that in the graph $\mathrm{Cin}(G, \pi_1)$, we have an edge between g_1 and g_2C_i. An analogous result holds for cells of the form $\{e\} \times C_i$. □

Note that the underlying Cayley graph of this Cayley incidence graph is the Cartesian product of the underlying Cayley graphs of $\mathrm{Cin}(G_1, \pi_1)$ and $\mathrm{Cin}(G_2, \pi_2)$.

In fact, Lemma 3.5 can let us construct non-trivial Cayley incidence graphs from trivial Cayley incidence graphs.

Example 3.6. Let G_1 and G_2 be two groups of the same order, possibly isomorphic, and consider the operation in Lemma 3.5 applied to $\mathrm{Cin}(G_1, \{G_1\})$ and $\mathrm{Cin}(G_2, \{G_2\})$. This gives us a Cayley incidence graph on $G_1 \times G_2$ with

$$\pi = \{G_1 \times \{e\}, \{e\} \times G_2\}.$$

Note that this is an instance of Example 2.5 since $G_1 \times \{e\}$ and $\{e\} \times G_2$ are subgroups of $G_1 \times G_2$.

Another way of constructing Cayley incidence graphs comes from taking intersections of two partitions.

Lemma 3.7. *Let* G *be a group and let* $\pi_1 = \{C_1, \ldots, C_\ell\}$ *and* $\pi_2 = \{C'_1, \ldots, C'_{\ell'}\}$ *be collections of subsets of* G *satisfying the* T*-axiom. Then the collection*

$$\pi_1 \cap \pi_2 := \{C \cap C' : C \in \pi_1, C' \in \pi_2\}$$

satisfies the T*-axiom.*

Proof. Clearly, each set in $\pi := \pi_1 \cap \pi_2$ contains the identity. Let $D \in \pi$ and $s \in D$. Then $D = C \cap C'$ for some $C \in \pi_1$ and $C' \in \pi_2$ and we have $s^{-1}C \in \pi_1$ and $s^{-1}C' \in \pi_2$. Therefore, $s^{-1}D = s^{-1}(C \cap C') = s^{-1}C \cap s^{-1}C' \in \pi$ as desired. □

Corollary 3.8. *Let* $\mathrm{Cin}(G, \pi)$ *and* $\mathrm{Cin}(G, \pi')$ *be two Cayley incidence graphs of* G*. For* $r \geq 1$*, define* π_r *by*

$$\pi_r = \{C_i \cap C'_j : C_i \in \pi, C'_j \in \pi', |C_i \cap C'_j| = r\}.$$

If π_r *is non-empty, then* $\mathrm{Cin}(G, \pi_r)$ *is a well-defined Cayley incidence graph of* G.

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Proof. Clearly, π_r is a collection of subsets of G of a fixed size r and they pairwise intersect precisely in the identity. By Lemma 3.7 and the fact that $|xC| = |C|$, for all $C \subseteq G$ and $x \in G$, we see that each non-empty π_r satisfies the T-axiom. Consequently, $\mathrm{Cin}(G, \pi_r)$ is a well-defined Cayley incidence graph. □

Example 3.9. Let $G = \mathbb{Z}_{15}$ and define

$$\pi = \{\{0,1,4,6\},\{0,2,11,12\},\{0,3,5,14\},\{0,9,10,13\}\},$$
$$\pi' = \{\{0,1,9,13\},\{0,2,3,11\},\{0,4,6,7\},\{0,8,12,14\}\}.$$

We see that $\pi_3 = \{\{0,2,11\},\{0,4,6\},\{0,9,13\}\}$ and $\mathrm{Cin}(G, \pi_3)$ is a 3-regular Cayley incidence graph of G.

4 Cayley hypergraphs

An important result in the theory of Cayley graphs is the following characterization of Sabidussi [50].

Theorem 4.1 (Sabidussi [50])**.** *A graph X is a Cayley graph if and only if it admits a group of automorphisms G acting regularly on the vertex set $V(X)$.*

This characterization motivates many of the definitions of Cayley hypergraphs that have been introduced over the years.

Teh and Shee [55] introduced the following hypergraph extension of Cayley graphs, called group hypergraphs. The definition is equivalent to the more recent reintroductions by Shee [52] and Lee and Kwon [40].

Definition 4.2 (Group Hypergraph)**.** Let G be a group and let $\mathcal{C}$ be a collection of non-empty subsets of G. The *group hypergraph* $\mathrm{CH}(G, \mathcal{C})$ is the hypergraph with vertex set G and hyperedge set

$$\{gC : g \in G, C \in \mathcal{C}\}.$$

Note that our Cayley incidence graphs $\mathrm{Cin}(G, \pi)$ are the incidence graphs of group hypergraphs $CH(G, \pi)$ with substantially more conditions placed on π.

Analogously to Theorem 4.1, Lee and Kwon [40] proved that a hypergraph is a group hypergraph if and only if it admits a group of automorphisms acting regularly on the vertex set. A similar result in the context of vertex transitive incidence structures was proven previously by Jajcay [37].

This led Jajcay and Jajcayová [38] to define the following version of Cayley hypergraphs.

Definition 4.3 (Uniform Cayley hypergraph)**.** A *uniform Cayley hypergraph* on vertex set V is an r-uniform hypergraph for some $1 \leq r \leq |V|$ that admits a group of automorphisms acting regularly on its vertices.

A more constructive version of Definition 4.3 comes from the restriction of Definition 4.2 to r-uniform hypergraphs. We can go one step further and ask for a version of Theorem 4.4 for uniform and regular hypergraphs.

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Theorem 4.4. *A hypergraph X is k-uniform and ℓ-regular with a group G acting regularly on the vertex set if and only if there exists a collection $\mathcal{C} = \{C_1, \ldots, C_\ell\}$ of subsets of G of size k that each contains the identity and satisfies the T-axiom, in which case X is isomorphic to $CH(G, \mathcal{C})$.*

Proof. We have already seen that all hypergraphs defined using a partition π satisfying the T-axiom are k-uniform and ℓ-regular. We may thus suppose X is a hypergraph admitting a group action which is regular on the vertex set.

By the variants of Sabidussi's theorem proved in [37, 40], we know that X is isomorphic to a group hypergraph $CH(G, \mathcal{C})$ for some collection $\mathcal{C}$. We may assume without loss of generality that $\mathcal{C}$ is the set of hyperedges that contain the identity.

Since X is ℓ-regular, we know that $|\mathcal{C}| = \ell$, and since X is k-uniform, every set $C \in \mathcal{C}$ must have size k. Finally, if $g \in C$ for some $C \in \mathcal{C}$, then $e \in g^{-1}C$, so $g^{-1}C \in \mathcal{C}$ and thus $\mathcal{C}$ satisfies the T-axiom. □

A hypergraph is *linear* if any two hyperedges intersect in at most one vertex or, equivalently, if its incidence graph has no four-cycles. From Lemma 3.4, we see that the hypergraphs arising from Cayley incidence graphs are linear. In fact, Cayley incidence graphs are precisely the incidence graphs of the linear hypergraphs of Theorem 4.4.

Corollary 4.5. *A linear hypergraph is uniform and regular with a group acting regularly on the vertex set if and only if its incidence graph is a Cayley incidence graph.*

Theorem 4.4 begs the question of why we would restrict ourselves to linear hypergraphs when, by placing less conditions on π, we get a larger class of uniform, regular hypergraphs satisfying a hypergraph version of Theorem 4.1.

Let $\mathcal{C} = \{C_1, \ldots, C_\ell\}$ be a collection of subsets of G such for all $1 \leq i \leq \ell$, the cell C_i has size k and contains the identity. If $\mathcal{C}$ satisfies the T-axiom, we can define X to be the bipartite incidence graph of $CH(G, \mathcal{C})$, that is, X has bipartition $\gamma = G$ and

$$\beta = \{gC : g \in G, C \in \mathcal{C}\}.$$

Note that

$$S(\mathcal{C}) = \bigcup_{C \in \mathcal{C}} C \setminus \{e\}$$

is still inverse-closed, so we can still speak about the underlying Cayley graph. However, the relationship between X and this underlying Cayley graph is weaker than the relationship for the Cayley incidence graphs. In the proof of Proposition 3.1, the key fact was that $g^{-1}h \neq e$ appears in a single cell of π if $g^{-1}h \in S$ and in no cells of π otherwise. For $\mathcal{C}$, there can be vertices in γ at distance two with a differing number of walks of length two between them. This has particularly devastating consequences for finding the spectrum, since we cannot determine it from the underlying Cayley graph.

We can remedy this problem by further requiring that every non-identity element that appears in $\mathcal{C}$ appears in some constant λ number of cells. Notice that this is not a consequence of the T-axiom.

Example 4.6. Let $G = \mathbb{Z}_7$, and let $C_1 = \{0, 1, 2\}$, $C_2 = \{0, 1, 6\}$, and $C_3 = \{0, 5, 6\}$. The set $\mathcal{C} = \{C_1, C_2, C_3\}$ satisfies the T-axiom, but the element 1 is contained in two cells, while the element 2 is only contained in one cell. In other words, as shown in Figure 2, vertices 0 and 1 share two common neighbours, while vertices 0 and 2 share a single common neighbour.

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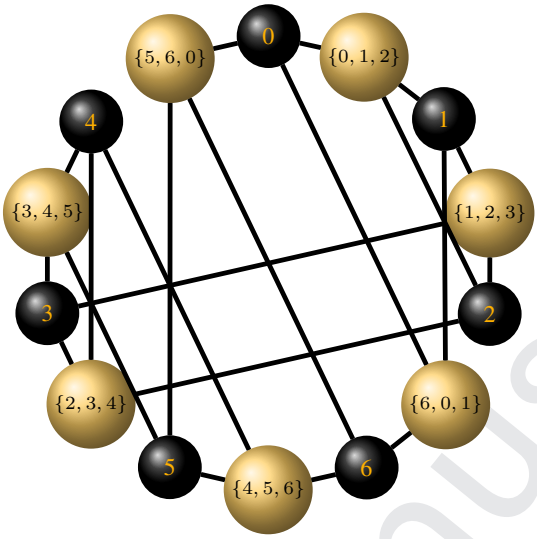


Figure 2: Incidence graph of $CH(\mathbb{Z}_7, \{\{0,1,2\},\{0,1,6\},\{0,5,6\}\})$

If we want to consider the class of graphs where the hypotheses of Theorem 4.4 are satisfied and an analogue of Proposition 3.1 is possible, we would need to restrict ourselves to hypergraphs where every pair of vertices is contained in either 0 or λ hyperedges. This is an unnatural constraint to place on hypergraphs unless $\lambda = 1$, in which case the hypergraphs are linear. Linear hypergraphs are relevant to yet another hypergraph extension of Cayley graphs which we discuss in Section 7 and some of our later results will depend on the linearity of the hypergraph.

Even though they were introduced in a very different context, the recent bipartite graphs of Evra, Feigon, Maurischat, and Parzanchevski [21] corresponds to a natural restriction of hypergraphs introduced by Shee [52], Lee and Kwon [40], and Jajcay and Jajcayová [38]. Hypergraphs give another perspective through which we can view Cayley incidence graphs, and this perspective will be particularly useful in Sections 7 and 8.

5 Designs and difference sets

Let J be the all-ones matrix. For a group G, $C \subseteq G$ and $g \in G$, define

$$Cg := \{sg : s \in C\}, \quad C^{-1} := \{s^{-1} : s \in C\}.$$

A 2-(v,k,λ) *design* (or a 2-*design*) is a point-block incidence structure with v points where every block contains k points and every pair of points is contained in λ blocks. Each point belongs to $r = \lambda(v-1)/(k-1)$ blocks, and the total number of blocks is $b = vr/k$. If N is the $v \times b$ incidence matrix, this definition is equivalent to the statement

$$NN^T = (r-\lambda)\, I + \lambda J.$$

A 2-design with $\lambda = 1$ is often referred to as a *Steiner system* $S(2,k,v)$. A *symmetric* 2-*design* is a 2-(v,k,λ) design with $b = v$, or equivalently, $k = r$. Symmetric 2-$(v,k,1)$

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designs are the point-line incidence structures of finite projective planes. Generally, constructions from finite geometry refer to lines instead of blocks, though for the purposes of this paper we will refer to them as blocks. For more background on designs, we refer the reader to the textbooks by Beth, Jungnickel, and Lenz [54] or Hughes and Piper [36].

Let $X = \mathrm{Cin}(G, \pi)$ be a Cayley incidence graph such that the underlying Cayley graph is complete, that is, $S(\pi) = G \setminus \{e\}$. Letting n be the order of G and k be the size of the cells of π, Proposition 3.1 tells us that X is the point-block incidence graph of a 2-$(n, k, 1)$ design. Note that G acts regularly on the points of this design. Conversely, by Corollary 4.5, the incidence graph of any 2-$(n, k, 1)$ design with an automorphism group acting regularly on its points is a Cayley incidence graph with a complete underlying Cayley graph.

When both halved graphs of a Cayley incidence graph $\mathrm{Cin}(G, \pi)$ are complete, we get the incidence graph of a projective plane with G acting regularly on its points. The Fano Plane of Example 2.7 is one such example. It is known [54] that the existence of a projective plane (a symmetric 2-$(n, k, 1)$ design) which has G as a point-regular automorphism group is equivalent to the existence of an $(n, k, 1)$-*difference set* of G. This implies a correspondence between these difference sets and Cayley incidence graphs with complete halved graphs, and we further show this correspondence in a more direct way.

Note that the only known examples of finite projective planes with a group acting regularly on their points are Desarguesian planes. It is a long standing open question [29] if there even exist non-Desarguesian finite projective planes with a point-transitive group action. In 1959, Ostrom and Wagner [49] proved that a finite projective plane with a group acting doubly transitively on its points is always Desarguesian.

Let G be a group of order n. A k-subset D of G is an (n, k, λ)-*difference set* if every non-identity element of G has exactly λ representations as a difference $d(d')^{-1}$ of two elements from D. The term "difference" is more natural in additive notation. We, however, continue to use multiplicative notation since the group G may be non-abelian. The *development* $\mathrm{dev}(D)$ of a difference set D is the incidence structure with point set G and block set $\{Dg : g \in G\}$. The development of an (n, k, λ)-difference set is a symmetric 2-(n, k, λ) design. More information on difference sets can be found in Beth, Jungnickel, and Lenz [54] or Jungnickel, Pott, and Smith [39].

Lemma 5.1. *Let G be a group, and let $D = \{d_1, \ldots, d_k\}$ be a k-subset of G. Then the following two conditions are equivalent:*

(1) *for all $1 \leq i, j \leq k$ and $i \neq j$, the elements $d_i d_j^{-1}$ are pairwise different;*

(2) *for all $1 \leq i, j \leq k$ and $i \neq j$, the elements $d_i^{-1} d_j$ are pairwise different.*

Proof. Suppose the first condition does not hold, i.e. $d_i d_j^{-1} = d_s d_t^{-1}$ for some $i \neq j$, $s \neq t$ and $(i, j) \neq (s, t)$. Note that $i = s$ would imply $j = t$, so we have $i \neq s$ and $j \neq t$. Then $d_s^{-1} d_i = d_t^{-1} d_j$ for $s \neq i$, $t \neq j$ and $(s, i) \neq (t, j)$, thus the second condition also does not hold. The same argument works in the other direction. □

If D is a difference set, then so is Dg for any $g \in G$, and, moreover, $\mathrm{dev}(D) = \mathrm{dev}(Dg)$. In particular, by possibly replacing D with Dd^{-1} for some $d \in D$, we may assume without loss of generality that a difference set D contains e.

Proposition 5.2. *Let G be a group of order n, let D be a k-subset of G such that $e \in D$, let $\pi_{D^{-1}} = \{dD^{-1} : d \in D\}$ and $\pi_D = \{d^{-1}D : d \in D\}$. Note that $\pi_{D^{-1}}$ and π_D satisfy the T-axiom. The following conditions are equivalent:*

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(1) *D is an $(n, k, 1)$-difference set;*

(2) *D^{-1} is an $(n, k, 1)$-difference set;*

(3) *Each non-identity element of G belongs to precisely one cell of $\pi_{D^{-1}}$;*

(4) *Each non-identity element of G belongs to precisely one cell of π_D.*

If these conditions hold, the point-block incidence graphs of $\mathrm{dev}(D)$ and $\mathrm{dev}(D^{-1})$ and the Cayley incidence graphs $\mathrm{Cin}(G, \pi_{D^{-1}})$ and $\mathrm{Cin}(G, \pi_D)$ are all isomorphic.

Proof. Conditions (1) and (3) hold when all elements $d(d')^{-1}$ for $d \neq d'$, $d, d' \in D$, are pairwise different, and conditions (2) and (4) hold when the same is true for elements $d^{-1}d'$. Thus they are all equivalent due to Lemma 5.1. The maps

$$\begin{cases} g \mapsto D^{-1}g \\ Dh \mapsto h \end{cases}, \quad \begin{cases} g \mapsto g^{-1} \\ Dh \mapsto h^{-1}D^{-1} \end{cases}, \quad \begin{cases} g \mapsto g^{-1}D \\ Dh \mapsto h^{-1} \end{cases}$$

define graph isomorphisms from the point-block incidence graph of $\mathrm{dev}(D)$ to the other three graphs. □

Note that in Proposition 5.2, points of $\mathrm{dev}(D)$ correspond to the blocks of $\mathrm{dev}(D^{-1})$.

The correspondence between difference sets and Cayley incidence graphs is more straightforward when G is abelian: if this is the case, $gD = Dg$ for any $g \in G$, so, continuing to use notation from Proposition 5.2, the point-block incidence graph of $\mathrm{dev}(D)$ is simply equal to $\mathrm{Cin}(G, \pi_D)$. The Fano Plane of Example 2.7 was constructed by $\mathrm{Cin}(\mathbb{Z}_7, \pi_D)$ for $D = \{0, 1, 3\}$.

Abel and Buratti [1] state that the existence of a 2-$(n, k, 1)$ design with an automorphism group acting regularly on its points is equivalent to the existence of what they call an $(n, k, 1)$-*partial difference family*. This implies a correspondence between these partial difference families and Cayley incidence graphs with complete underlying Cayley graphs. This can also be shown more directly in a similar manner to Proposition 5.2, though we omit the details.

We conclude this section by showing how finite projective and affine spaces (Desarguesian projective and affine planes in the case of dimension 2) can be constructed as Cayley incidence graphs. Both of these constructions can be restated in terms of partial difference families.

For a prime power q, denote by $\mathbb{F}_q$, $\mathbb{F}_q^*$ and $\mathbb{F}_q^+$ the finite field of size q, its multiplicative group and its additive group, respectively.

Let $n \geq 2$, let q be a prime power, and let α be a generator of $\mathbb{F}_{q^n}^*$. Since α is a root of some primitive polynomial $f(x)$ of degree n in $\mathbb{F}_q$, every element of $\mathbb{F}_{q^n}$ can be written as a sum,

$$\sum_{i=0}^{n-1} c_i \alpha^i, \quad c_i \in \mathbb{F}_q,$$

thus giving a representation of $\mathbb{F}_{q^n}$ as an n-dimensional vector space over $\mathbb{F}_q$. Denote by $\mathcal{V}_1$ the set of all 1-dimensional subspaces of this vector space, i.e. sets of the form $\langle a \rangle = \{ka : k \in \mathbb{F}_q\}$ for some $a \in \mathbb{F}_{q^n} \setminus \{0\}$. Let $[n]_q = |\mathcal{V}_1| = (q^n - 1)/(q - 1)$.

The following example for prime q appears in Abel and Buratti [1].

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Example 5.3. Let $G := (\mathbb{F}_q^+)^n$ and $\pi := \mathcal{V}_1$. Note that for any two different $\langle a\rangle, \langle b\rangle \in \pi$ we have $\langle a\rangle \cap \langle b\rangle = \{0\}$. Also, each $\langle a\rangle \in \pi$ is an additive subgroup of G, so π trivially satisfies the T-axiom. The Cayley incidence graph $\mathrm{Cin}(G,\pi) = (\gamma \cup \beta, E)$ is in fact the point-line incidence graph of the affine space $\mathrm{AG}(n,q)$ with points γ, lines β, and π corresponding to the set of lines passing through the origin. Note that this construction is a particular instance of Example 2.5 since π is a collection of subgroups with pairwise trivial intersections.

The next example for $n = 3$ corresponds to Singer difference sets [53].

Example 5.4. Let $n \geq 3$. Denote by $\mathcal{V}_2$ the set of all 2-dimensional subspaces of $\mathbb{F}_{q^n}$ as a vector space over $\mathbb{F}_q$. Since $\mathbb{F}_q^*$ is a subgroup of $\mathbb{F}_{q^n}^*$, we have $\alpha^k \in \mathbb{F}_q^*$ if and only if k is divisible by $\left|\mathbb{F}_{q^n}^*\right| / \left|\mathbb{F}_q^*\right| = [n]_q$. In particular, $\langle \alpha^i\rangle \in \mathcal{V}_1$ are pairwise different for $0 \leq i < [n]_q$, so $\mathcal{V}_1 = \{\langle \alpha^i\rangle : 0 \leq i < [n]_q\}$. Now $\mathcal{V}_1$ may be identified with $G := \mathbb{Z}_{[n]_q}$, and $\mathcal{V}_2$ may be identified with certain $(q+1)$-subsets of G.

Let π be the collection of $(q+1)$-subsets of G corresponding to 2-dimensional subspaces containing $\langle \alpha^0\rangle$, i.e. sets of the form

$$\langle \alpha^0, \alpha^i\rangle = \left\{k_1\alpha^0 + k_2\alpha^i : k_1, k_2 \in \mathbb{F}_q\right\} \in \mathcal{V}_2$$

for some $0 < i < [n]_q$. By construction, each $C \in \pi$ contains 0. For two non-equal $\langle \alpha^0, \alpha^i\rangle, \langle \alpha^0, \alpha^j\rangle \in \mathcal{V}_2$ we have $\langle \alpha^0, \alpha^i\rangle \cap \langle \alpha^0, \alpha^j\rangle = \langle \alpha^0\rangle$, so two different cells of π have only 0 in common. Moreover, π satisfies the T-axiom: if $C \in \pi$, $s \in C \setminus \{0\}$, then C corresponds to $\langle \alpha^0, \alpha^s\rangle$, thus $C - s$ corresponds to $\alpha^{-s} \cdot \langle \alpha^0, \alpha^s\rangle = \langle \alpha^{-s}, \alpha^0\rangle \in \mathcal{V}_2$, so $C - s \in \pi$. The Cayley incidence graph $\mathrm{Cin}(G,\pi)$ is the incidence graph for the inclusion relation between $\mathcal{V}_1$ and $\mathcal{V}_2$, so it is the point-line incidence graph of the projective space $\mathrm{PG}(n-1,q)$.

The collection π described above can be calculated explicitly if the polynomial $f(x)$ is known; for example, to get Fano Plane of Example 2.7 we can take $n = 3$, $q = 2$ and $f(x) = x^3 - x - 1$.

6 β-transitive Cayley incidence graphs

From Corollary 4.5, we know that for a Cayley incidence graph $\mathrm{Cin}(G,\pi) = (\gamma \cup \beta, E)$ the action of G on γ is regular. However, this action need not be regular, or even transitive, on β.

Example 6.1. The graphs in Example 3.6 clearly have two orbits on β under $G_1 \times G_2$.

For any Cayley incidence graph $\mathrm{Cin}(G,\pi)$ with vertex set $\gamma \cup \beta$, we will show that the orbits under the action of G on β are classified by a relation on π. Then we will consider when the action is regular within each orbit.

Define $gC_i \approx hC_j$ if and only if gC_i and hC_j belong to the same orbit under G. This is an equivalence relation, and

$$\begin{aligned} C_i \approx C_j &\iff \exists g \in G : gC_j = C_i \\ &\iff \exists g \in C_i : gC_j = C_i \\ &\iff \exists g \in C_i : C_j = g^{-1}C_i, \end{aligned}$$

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so $C_i \approx C_j$ if and only if there is a $g \in C_i$ such that $g^{-1}C_i = C_j$.

Since $gC_i \approx C_i$ for any $g \in G$, we see that $gC_i \approx hC_j$ if and only if $C_i \approx C_j$. We then restrict the relation $\approx$ to $\pi = \{C_1, \ldots, C_\ell\}$, since this contains the same information as the full relation on β.

Denote by $[C]$ the equivalence class of $C \in \pi$ under the relation $\approx$, that is

$$[C] = \{s^{-1}C \ : \ s \in C\}.$$

Definition 6.2. A Cayley incidence graph $\mathrm{Cin}(G, \pi)$ is called *β-transitive* if for every $C_i, C_j \in \pi$ there is some $s \in C_i$ such that $s^{-1}C_i = C_j$. Equivalently, $\mathrm{Cin}(G, \pi)$ is β-transitive if $[C] = \pi$ for some $C \in \pi$.

Note that this is not equivalent to $\mathrm{Cin}(G, \pi)$ having at most two orbits on the vertices. By our definition, β-transitivity refers to the action of G on β, not the action of the full automorphism group.

Example 6.3. The graph in Example 2.7 (Fano Plane) is β-transitive. More generally, the Cayley incidence graphs coming from the difference set construction in Proposition 5.2 are β-transitive.

Let $X = \mathrm{Cin}(G, \pi)$ be a Cayley incidence graph and let $C \in \pi$. If X is β-transitive, then $[C] = \pi$. More generally, $[C]$ satisfies the T-axiom, so $\mathrm{Cin}(G, [C])$ is an induced subgraph of X on which G acts transitively. This leads to the following result.

Proposition 6.4. *Let $\mathrm{Cin}(G, \pi) = (\gamma \cup \beta, E)$ be a Cayley incidence graph and let π consist of t equivalence classes: $\pi = [C_1] \sqcup \cdots \sqcup [C_t]$. The orbits of elements of β under the action of G are precisely*

$$\beta_i := \{gC_i : g \in G\}, 1 \leq i \leq t.$$

This allows us to decompose a Cayley incidence graph into β-transitive Cayley incidence graphs. We cannot do the same if we weaken the linearity assumption: let $\mathcal{C} = \{C_1, \ldots, C_\ell\}$ such that for all $1 \leq i \leq \ell$ the cell C_i has k elements and contains the identity. Suppose further that $\mathcal{C}$ satisfies the T-axiom and that every non-identity element in $\bigcup_{i=1}^{\ell} C_i$ lies in the same number of cells of $\mathcal{C}$. We can still define β-transitive exactly as in Definition 6.2, but in a non-β-transitive case the equivalence class $[C_i]$ might not preserve the property that non-identity elements appear in the same number of cells, as can be seen in the following example.

Example 6.5. Let $G = \mathbb{Z}_6$ and $\mathcal{C} = \{\{0, 1, 2\}, \{0, 1, 5\}, \{0, 4, 5\}, \{0, 2, 4\}\}$. Then $\mathcal{C}$ satisfies the T-axiom and each non-identity element in $\bigcup_{C \in \mathcal{C}} C$ lies in 2 cells of $\mathcal{C}$. However, the same does not hold for the equivalence class $[\{0, 1, 2\}] = \{\{0, 1, 2\}, \{0, 1, 5\}, \{0, 4, 5\}\}$: 1 and 5 appear in two cells, while 2 and 4 appear only in one cell.

Now for any $gC \in \beta$ we will compute its *setwise stabilizer*

$$\mathrm{Stab}_G(gC) := \{x \in G : x(gC) = gC\}.$$

Since $\mathrm{Stab}_G(gC) = g\,\mathrm{Stab}_G(C)g^{-1}$, it suffices to compute $\mathrm{Stab}_G(C)$ for $C \in \pi$.

Proposition 6.6. *Let $\mathrm{Cin}(G, \pi)$ be a Cayley incidence graph. For any $C \in \pi$ we have*

$$\mathrm{Stab}_G(C) = C \cap C^{-1}.$$

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Proof. Let $S := C \cap C^{-1}$; our goal is to prove that $\mathrm{Stab}_G(C) = S$. For any $s \in S$, we have $sC \in \pi$ and $s \in sC \cap C$, thus $sC = C$, so $s \in \mathrm{Stab}_G(C)$. Conversely, let $g \in \mathrm{Stab}_G(C)$, then $gC = C$, and since $e \in C$, it follows that $g, g^{-1} \in C$, thus $g \in S$. □

Corollary 6.7. *Let $X := \mathrm{Cin}(G, \pi) = (\gamma \cup \beta, E)$ be a Cayley incidence graph. The action of G on β is regular if and only if X is β-transitive and $\mathrm{Stab}_G(C) = \{e\}$ for some $C \in \pi$.*

When the action of G on β is regular, we will call it *β-regular*.

Remark 6.8. A β-transitive graph $\mathrm{Cin}(G, \pi)$ always has $l \leq k$ because $\pi = [C]$ for any $C \in \pi$ and so $\ell = |\pi| = |[C]| \leq |C| = k$. Equality holds exactly when $\mathrm{Stab}_G(C) = \{e\}$, so β-regular graphs are always regular as graphs.

A *Dedekind group* is a group in which every subgroup is normal. In particular, abelian groups are Dedekind groups.

Corollary 6.9. *Let $\mathrm{Cin}(G, \pi)$ be a Cayley incidence graph for some Dedekind group G. For any $C \in \pi$, either $\mathrm{Stab}_G(C) = C$ or $\mathrm{Stab}_G(C) = \{e\}$.*

Proof. Suppose that $\mathrm{Stab}_G(C) \neq C$, so by Proposition 6.6, there exists some $s \in C$ with $s^{-1} \notin C$. By the T-axiom, $C' := s^{-1}C \in \pi$, and $s^{-1} \notin C$ implies $C \neq C'$. Since $\mathrm{Stab}_G(C)$ is a normal subgroup of G, we have

$$\mathrm{Stab}_G(C') = s^{-1}\,\mathrm{Stab}_G(C)s = \mathrm{Stab}_G(C).$$

But then $\mathrm{Stab}_G(C) \subseteq C \cap C' = \{e\}$. □

We conclude this section with a few more results on Cayley incidence graphs for Dedekind and abelian groups.

Lemma 6.10. *Let G be a Dedekind group, and let $\mathrm{Cin}(G, \pi)$ be an (ℓ, k)-biregular β-transitive Cayley incidence graph. Then $\ell = 1$ or $\ell = k$.*

Proof. Fix any $C \in \pi$, then $\pi = [C] = \{s^{-1}C : s \in C\}$. If $\ell \neq k$, then, by Remark 6.8, $\ell < k$ and $\mathrm{Stab}_G(C) \neq \{e\}$, so by Corollary 6.9, $\mathrm{Stab}_G(C) = C$. Thus C is a subgroup, which implies $s^{-1}C = C$ for every $s \in C$. Since $\pi = [C] = \{C\}$, we must have $\ell = 1$. □

Lemma 6.11. *Let $X = \mathrm{Cin}(G, \pi)$ be an (ℓ, k)-biregular β-transitive Cayley incidence graph with G abelian. If $\ell \geq 3$, then X has girth six.*

Proof. By Lemma 3.4, X has girth at least six. Let $C_1, C_2, C_3 \in \pi$ be pairwise distinct. Since $\mathrm{Cin}(G, \pi)$ is β-transitive, there are some $s \in C_1$, $t \in C_2$ such that $C_2 = s^{-1}C_1$ and $C_3 = t^{-1}C_2$. We now show that

$$(e, C_1, s, sC_3, t^{-1}, C_3, e)$$

is a 6-cycle in X. It is clear that $e \in C_1, C_3$, so $s \in sC_3$, and by assumption $s \in C_1$, so we only need to show that $t^{-1} \in C_3, sC_3$. We have just noted that $C_3 = t^{-1}C_2$, which implies $t^{-1} \in C_3$ since $e \in C_2$. We also have

$$sC_3 = st^{-1}C_2 = t^{-1}C_1$$

since G is abelian. Thus $t^{-1} \in sC_3$. □

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Note that the same result does not hold if G is non-abelian.

Example 6.12. Let G be the non-abelian group of order 21. It is the semidirect product of $\mathbb{Z}_7$ and $\mathbb{Z}_3$ so we can write $G = \langle a, b\rangle$ where a has order seven and b has order three. Let $\pi = \{C_1, C_2, C_3\}$ with

$$\begin{aligned} C_1 &= \{e, b, b^{-1}\}, \\ C_2 &= \{e, ab, (ab)^{-1}\} \\ C_3 &= \{e, a^2b, (a^2b)^{-1}\} \end{aligned}$$

Then π satisfies the T-axiom and $k = \ell = 3$. The Cayley incidence graph is regular, but not a Cayley graph, and it has girth eight.

7 Subdivision graphs and cages

Whenever we have a bipartite biregular graph where one bipart has valency two, we can consider it as the subdivision graph of a regular graph. We used this in Example 2.6 to show that the subdivision graph of any Cayley graph is a Cayley incidence graph.

However, if π has two cells, then the Cayley incidence graph is the subdivision graph induced by the other bipart and not the underlying Cayley graph. Thus we wish to characterize the Cayley incidence graphs $\mathrm{Cin}(G, \{C_1, C_2\})$.

Proposition 7.1. *Let $X = \mathrm{Cin}(G, \{C_1, C_2\})$ be a Cayley incidence graph. Then there exist subgroups S_1, S_2 with trivial intersection such that either:*

(1) *$C_1 = S_1$ and $C_2 = S_2$;*

(2) *there exists some non-identity group element x such that $S_2 = x^{-1}S_1x$, $C_1 = S_1 \cup S_1x$ and $C_2 = S_2 \cup S_2x^{-1}$.*

Proof. If X is not β-transitive, then clearly (1) holds. Thus we may assume there exists some $x \in C_1$ such that $x^{-1}C_1 = C_2$. Let $S_1 = \mathrm{Stab}_G(C_1)$ and observe that $S_1x \subseteq C_1$. To see that these are all the elements of C_1, we suppose that there is some other element y such that $y^{-1}C_1 = C_2$. Then $yx^{-1} \in S_1$ since $yx^{-1}C_1 = yC_2 = C_1$, and thus $y \in S_1x$. Similarly, letting $S_2 = \mathrm{Stab}_G(C_2)$, we get $C_2 = S_2 \cup S_2x^{-1}$, and by the property of stabilizers, $C_2 = x^{-1}C_1$ implies $S_2 = x^{-1}S_1x$. □

For abelian groups, this gives us a very restricted form.

Corollary 7.2. *Let G be an abelian group of order n and let $X = \mathrm{Cin}(G, \{C_1, C_2\})$ be a connected $(2, k)$-biregular Cayley incidence graph. Then either $k = 2$ and X is a cycle, or there exists some m such that $n = mk$ and X is the subdivision graph of the complete bipartite graph $K_{m,m}$.*

Proof. By Corollary 6.9, the only way Case (2) of Proposition 7.1 can hold is if $S_1 = S_2 = \{e\}$, in which case we have $\mathrm{Cin}\big(G, \big\{\{e, x\}, \{e, x^{-1}\}\big\}\big)$, which is a cycle.

Otherwise, C_1 and C_2 are two disjoint subgroups of order k. Then the halved graph H_β is the bipartite graph with biparts given by the cosets G/C_1 and G/C_2. Since the elements of $C_1 \cup C_2$ generate G, we must have $G \cong C_1 \times C_2$. Thus

$$m := |G/C_1| = |C_1| = |C_2| = |G/C_2|$$

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is both the valency of vertices in H_β and the sizes of the biparts. It follows that H_β is the complete bipartite graph $K_{m,m}$ and X is its subdivision graph. □

For non-abelian groups, the examples are less trivial. Given two subgroups S_1, S_2, the *rank-two coset geometry* can be viewed as a bipartite graph with biparts formed from the cosets of G/S_1 and G/S_2. The classification of coset geometries was spearheaded by Buekenhout [7, 8].

Many results in this area are concerned with classifying rank-two coset geometries for a particular group. The incidence graph of a rank-two coset geometry is always bipartite and edge-transitive. De Saedeleer, Leemans, Mixer, and Pisanki [13] took the opposite approach and studied regular, edge-transitive bipartite graphs of valency three and four to see which graphs were the incidence graphs of rank-two coset geometries.

Case (1) in Proposition 7.1 is the subdivision graph of the incidence graph of a rank-two coset geometry with $|S_1| = |S_2|$ and S_1 and S_2 intersecting only in the identity.

Example 7.3. Let G be the unique non-abelian group of order 21, that is $G \cong \mathbb{Z}_7 \rtimes \mathbb{Z}_3$. Let a and b be generators of order three and seven, respectively and define $C_1 := \langle a \rangle$ and $C_2 := \langle ab \rangle$. Then C_1 and C_2 are subgroups of order three with trivial intersection and the graph $\mathrm{Cin}(G, \{C_1, C_2\})$ is a $(2, 3)$-biregular graph of girth twelve. The halved graph H_β is isomorphic to the Heawood graph, shown in Figure 1.

We also have non-trivial examples of Case (2) in Proposition 7.1.

Example 7.4. Consider $G = S_4$, and let $C_1 = \{e, (12), (134), (1342)\}$ and $C_2 = \{e, (24), (431), (1243)\}$. This gives an example of Case (2) where $x_1 = (134)$ and $S_1 = \{e, (12)\}, S_2 = \{e, (24)\}$.

A (k, g)-*cage* is a graph with the minimum number of vertices over all k-regular graphs of girth g. We can similarly define an (ℓ, k, g)-biregular cage as a bipartite graph with minimum number of vertices over all (ℓ, k)-biregular graphs of girth g. This is equivalent to viewing them as hypergraph cages as done by Ellis and Linial [18] or Erskine and Tuite [19]. More information about regular cages can be found in the survey of Exoo [22], but one major idea is that Cayley graphs can be used to construct graphs of given valency and large girth.

We could use Proposition 7.1 to develop a strategy to search for $(2, 3, g)$ or $(2, 4, g)$-cages using Cayley incidence graphs.

In Case (1), the work of De Saedeleer et al. [13] gives coset geometries with 3- and 4-regular incidence graphs. Using the subgroups from these coset geometries as the cells of our Cayley incidence graphs gives us $(2, 3)$ and $(2, 4)$–biregular graphs. Thus, if we begin by looking at cubic and quartic bipartite edge-transitive graphs with large girth, we can construct $(2, 3)$ and $(2, 4)$-biregular graphs with large girth.

In Case (2), we can build the cells of the Cayley incidence graph in a deliberate way from the groups. By Lemma 6.11, we know we can restrict our search to non-abelian groups, and then from Proposition 7.1 we look for nontrivial subgroups of order $\frac{k}{2}$ in the same conjugacy class. This cannot happen when $k = 3$, and when $k = 4$ the desired subgroups have order 2.

This would be a computationally viable, but mathematically inefficient, way to search for $(2, k)$-biregular Cayley incidence graphs of large girth. One of the steps involved is to look at k-regular graphs with large girth and impose additional restrictions for which graphs

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we can construct as Cayley incidence graphs. More generally, for any k-regular graph with large girth, we can construct a $(2,k)$-biregular bipartite graph with twice the girth simply by subdividing every edge once. However, a similar strategy for $(3,k)$-biregular Cayley incidence graphs should lead to non-trivial constructions.

This is the general approach used by Erskine and Tuite [19] and Ellis and Linial [18] in considering hypergraph cages. However, rather than using Cayley incidence graphs, they used another hypergraph generalization of Cayley graphs due to Buratti [9].

Definition 7.5 (t-Cayley hypergraph)**.** Let G be a group, let S be a subset of G that does not contain the identity, and let t be an integer between two and the maximum order of elements of S. Then the *t-Cayley hypergraph* has vertex set G and hyperedge set

$$\left\{gs^i : 0 \leq i \leq t-1, g \in G, s \in S\right\}.$$

For a set $S \subseteq G \setminus \{e\}$, we can define a collection of subsets

$$\mathcal{C} := \left\{\left\{e, s, s^2, \ldots, s^{t-1}\right\} : s \in S\right\},$$

from which it is clear that t-Cayley hypergraphs are a subclass of the more general class of group hypergraphs in Definition 4.2. If t is at most the minimum order of elements of S, then the t-Cayley hypergraph is t-uniform.

Buratti [9] verified that a t-Cayley hypergraph has a group of automorphisms which act regularly on the vertices, but also showed that the Fano plane is a 3-uniform hypergraph which has a group of automorphisms acting regularly on the vertices, even though it is not a 3-Cayley hypergraph. Notably, the Fano Plane is the Cayley incidence graph of Example 2.7 and a $(3,6)$-cage, or $(3,3,6)$-biregular cage.

Ellis and Linial [18] independently defined a construction of random Cayley hypergraphs that is a subconstruction of t-Cayley hypergraphs. In the notation of Buratti [9], Ellis and Linial [18] let G be the symmetric group and let S be a set chosen uniformly at random from permutations with a particular structure. They proved that with high probability, the ensuing t-Cayley hypergraph is a linear, uniform, regular hypergraph with large girth.

A related, but more explicit, construction using t-Cayley hypergraphs was given recently by Erskine and Tuite [19], who used Definition 7.5 to obtain computational results improving the upper bound on the size of hypergraph cages. We propose that Cayley incidence graphs are precisely the right class of graphs to search for biregular cages. The restriction of Definition 4.2 to t-Cayley graphs excludes important graphs, like the Fano Plane, but by Corollary 4.5 we see that the Cayley incidence graphs are precisely the restriction of Definition 4.2 that might give rise to biregular cages.

8 Automorphisms

We have already seen how a group G acts on $\mathrm{Cin}(G,\pi)$. In this section we will further investigate the full automorphism group of a Cayley incidence graph by considering how group automorphisms may yield additional automorphisms of the graph.

One notable use of Cayley graphs was by Frucht [26] to prove that every group is the automorphism group of a finite undirected graph. The idea of the proof and subsequent strengthenings was to begin with a Cayley graph and replace edges with certain subgraphs to disrupt the extra symmetries and obtain the desired group. This is related to the problem of *graphical regular representations*, i.e. Cayley graphs with $\mathrm{Aut}(\mathrm{Cay}(G,S)) \cong G$.

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Godsil [30] proved that with the exception of two infinite families and 13 simple groups with order at most 32, every group admits a graphical regular representation. Babai, Godsil, Imrich, and Lovasz [5] conjectured that if (G_n) is a family of groups with order n tending to infinity that is neither abelian with exponent greater than two nor generalized dicyclic, then almost all Cayley graphs are graphical regular representations. This conjecture was proven recently by Xia and Zheng [58].

When considering the automorphism group of a Cayley incidence graph $X = \mathrm{Cin}(G, \pi) = (\gamma \cup \beta, E)$, there is some choice to be made. A graph automorphism maps $\gamma \cup \beta$ to $\gamma \cup \beta$ and preserves the adjacency relation between γ and β. However, a hypergraph automorphism maps γ to itself and preserves the hyperedge relations. In particular, if a Cayley incidence graph is regular, we might have a graph automorphism mapping γ to β which is not a hypergraph automorphism.

For any Cayley incidence graph we have $\mathrm{Aut}(CH(G, \pi)) \subseteq \mathrm{Aut}(\mathrm{Cin}(G, \pi))$, and so we will begin by considering the hypergraph automorphism group.

Regular representations of hypergraphs were considered by Bayat, Alaeiyan, and Firouzian [6] for t-Cayley hypergraphs, and by Jajcay [37] for group hypergraphs. We are specifically interested in Cayley incidence graphs and a result used by Godsil [31] to prove that almost all Cayley graphs of certain groups are graphical regular representations.

Given a graph X, and a group G acting transitively on the vertices of X, Godsil [31] determined the normalizer of G in $\mathrm{Aut}(X)$. We similarly determine the normalizer of the group G in the automorphism group of a Cayley incidence graph.

We say that a group automorphism $\varphi \in \mathrm{Aut}(G)$, *induces a permutation of* $\pi = \{C_1, \ldots, C_\ell\}$, if for each $1 \leq i \leq \ell$ there exists some $j \in \{1, \ldots, \ell\}$ such that $\varphi(C_i) = C_j$. Let

$$\mathrm{Aut}(G, \pi) := \{\varphi \in \mathrm{Aut}(G) : \varphi \text{ induces a permutation of } \pi\}.$$

For a group G and a subgroup K, we will denote by $N_G(K)$ the normalizer of K in G. Let $AH = \mathrm{Aut}(CH(G, \pi))$. For $g \in G$, let $\mathcal{L}_g \in \mathrm{Aut}(G)$ be the left translation $\mathcal{L}_g : G \to G : x \mapsto gx$. The set $\{\mathcal{L}_g : g \in G\}$ forms a subgroup of $\mathrm{Aut}(G)$ isomorphic to G. We will abuse the notation and refer to this subgroup as G. The following result is known, see for example [14, Corollary 4.2B].

Lemma 8.1. *Let G be a group, and let $f : G \to G$ be a bijective function with $f(e) = e$, such that for all $g \in G$,*

$$f \circ \mathcal{L}_g \circ f^{-1} = \mathcal{L}_h$$

for some $h \in G$. Then f is a group automorphism.

Proof. We see that $f(g) = f(gf^{-1}(e)) = h$. Thus, we have

$$f \circ \mathcal{L}_g \circ f^{-1} = \mathcal{L}_{f(g)},$$

and therefore,

$$\mathcal{L}_{f(gh)} = f \circ \mathcal{L}_{gh} \circ f^{-1} = f \circ \mathcal{L}_g \circ f^{-1} \circ f \circ \mathcal{L}_h \circ f^{-1} = \mathcal{L}_{f(g)} \circ \mathcal{L}_{f(h)}.$$

This implies $\mathcal{L}_{f(gh)}(e) = \mathcal{L}_{f(g)} \circ \mathcal{L}_{f(h)}(e)$, so we have $f(gh) = f(g)f(h)$. Then, since $f(e) = e$, f is a group automorphism. □

Theorem 8.2. *Let $\mathrm{Cin}(G, \pi)$ be a Cayley incidence graph. Then*

$$N_{AH}(G) \cap \{\varphi \in AH : \varphi(e) = e\} \cong \mathrm{Aut}(G, \pi).$$

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Proof. We recall that the vertex set of $CH(G,\pi)$ is given by G. Hence, hypergraph automorphisms $f \in N_{AH}(G)$ are in particular bijections $f\colon G \to G$, and G is a subgroup of $\mathrm{Aut}(CH(G,\pi))$.

Let $f\colon G \to G$ be a hypergraph automorphism with $f \in N_{AH}(G)$ and $f(e) = e$. Since $f \in N_{AH}(G)$, for every $g \in G$ there exists some $h \in G$ such that

$$f \circ \mathcal{L}_g \circ f^{-1} = \mathcal{L}_h.$$

Thus, by Lemma 8.1, f is a group automorphism. Since $f(e) = e$, we also see that f must induce a permutation of π, by the fact that it is a hypergraph automorphism. Hence $f \in \mathrm{Aut}(G,\pi)$.

We now show that any $\varphi \in \mathrm{Aut}(G,\pi)$ defines a hypergraph automorphism and that $\varphi \in N_{AH}(G)$. Let gC_i be a hyperedge. Then $\varphi(gC_i) = \varphi(g)\varphi(C_i)$ and hence the result is a hyperedge. Since $\mathrm{Aut}(G,\pi)$ is a group, we see that the preimage of any hyperedge is also a hyperedge. Thus elements of $\varphi \in \mathrm{Aut}(G,\pi)$ define automorphisms of $\mathrm{CH}(G,\pi)$.

For any $h \in G$, we have

$$\varphi \circ \mathcal{L}_g \circ \varphi^{-1}(h) = \varphi(g)h,$$

and hence $\varphi \circ \mathcal{L}_g \circ \varphi^{-1} = \mathcal{L}_{\varphi(g)}$. Thus $\varphi \in N_{AH}(G)$. □

Remark 8.3. Theorem 8.2 has a straightforward generalization to the automorphism groups of group hypergraphs. The proof is identical.

Corollary 8.4. *Let* $\mathrm{Cin}(G,\pi)$ *be a Cayley incidence graph. Then*

$$N_{AH}(G) \cong G \rtimes \mathrm{Aut}(G,\pi).$$

Proof. Let $f \in N_{AH}(G)$. Suppose that $f(e) = g$. Then $\mathcal{L}_{g^{-1}} \circ f(e) = e$, and thus $\mathcal{L}_{g^{-1}} \circ f \in \mathrm{Aut}(G,\pi)$ by the previous proposition. It is now clear that any element of $N_{AH}(G)$ can be written as a product of $g \in G$ and $f \in \mathrm{Aut}(G,\pi)$, and it is clear that $G \cap \mathrm{Aut}(G,\pi) = \{e\}$. Thus $N_{AH}(G)$ is the stated semi-direct product. □

Now let us consider the existence of automorphisms which swap the two sides of $\mathrm{Cin}(G,\pi)$. In terms of hypergraphs, we can think of these as isomorphisms between a hypergraph and its dual. However, these are truly automorphisms of the bipartite graph, and it will be convenient to therefore view them as graph automorphisms with $A = \mathrm{Aut}(\mathrm{Cin}(G,\pi))$. We note that one can only have an automorphism $f \in N_A(G)$ which swaps γ and β if the graph is β-regular, so we restrict our attention to this case. As indicated earlier, any hypergraph automorphism $\varphi \in AH$ induces an automorphism of the Cayley incidence graph $\mathrm{Cin}(G,\pi)$, given by

$$f_\varphi : \gamma \cup \beta \to \gamma \cup \beta : \begin{cases} x \mapsto \varphi(x) & \text{if } x \in \gamma \\ gC \mapsto \varphi(g)\varphi(C) & \text{if } gC \in \beta. \end{cases}$$

This defines an injective group homomorphism

$$\theta : AH \to A : \varphi \mapsto f_\varphi,$$

and it allows one to identify AH as a subgroup of A. In what follows, we will abuse notation, and simply identify $\theta(AH)$ with this subgroup of A.

To understand when an automorphism can swap two sides of a Cayley incidence graph, we introduce the following construction.

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Proposition 8.5. *Let $X = \mathrm{Cin}(G, \pi)$ be β-regular with $\pi = \{C_1, C_2, \ldots, C_k\}$, $C_1 = \{g_1, g_2, \ldots, g_k\}$, $g_1 = e$ and $C_i = g_i^{-1}C_1$. Let $\pi^* = \{C_1^{-1}, g_2C_1^{-1}, \ldots, g_kC_1^{-1}\}$. Then $\mathrm{Cin}(G, \pi^*)$ is a well-defined Cayley incidence graph isomorphic to X.*

Proof. The cells $C_1^*, \ldots C_k^*$ of π^* are given by

$$\begin{aligned} C_1^* &= \{e, g_2^{-1}, \ldots, g_k^{-1}\} \\ C_2^* &= \{g_2, e, \ldots, g_2g_k^{-1}\} \\ &\vdots \\ C_k^* &= \{g_k, g_kg_2^{-1}, \ldots, g_kg_{k-1}^{-1}, e\}. \end{aligned}$$

It is straightforward to check that π^* satisfies the T-axiom. We additionally need to check that the cells $C_i^* = \{g_ig_j^{-1} : g_j \in C_1\}$ only intersect in the identity. Since $C_i = g_i^{-1}C_1 = \{g_i^{-1}g_j : g_j \in C_1\}$, and since two cells $C_i, C_a \in \pi$ only intersect in the identity, we see that the elements $g_i^{-1}g_j$ are all pairwise distinct when $i \neq j$. Hence, by Lemma 5.1, the elements $g_ig_j^{-1}$ are all distinct. Thus two cells C_i^*, C_a^* only intersect in the identity, and hence $\mathrm{Cin}(G, \pi^*)$ is well defined.

Let $\gamma \cup \beta$ and $\gamma^* \cup \beta^*$ be the vertex sets of $\mathrm{Cin}(G, \pi)$ and $\mathrm{Cin}(G, \pi^*)$, respectively. Let ψ be the map $\gamma \cup \beta \to \gamma^* \cup \beta^*$ defined by

$$\begin{cases} g \mapsto gC_1^* \text{ for } g \in \gamma \\ gC_1 \mapsto g \text{ for } gC_1 \in \beta, \end{cases}$$

Note that $h \in gC_1$ if and only if $\psi(gC_1) = g \in hC_1^* = \psi(h)$, thus ψ is a graph homomorphism. Since $(\pi^*)^* = \pi$, it is easy to see that

$$\varphi\colon G \to G : \begin{cases} g \mapsto gC_1 \text{ for } g \in \gamma^* \\ gC_1^* \mapsto g \text{ for } g \in \beta^*, \end{cases}$$

defines a two-sided inverse of ψ. □

Corollary 8.6. *Let $X = \mathrm{Cin}(G, \pi)$ be a β-regular Cayley incidence graph. Then both halved graphs of X are Cayley graphs and if $S(\pi) = S(\pi^*)$, the two halved graphs are isomorphic.*

Note that for an abelian group $S(\pi) = S(\pi^*)$ always holds.

We will need the following standard result about bipartite graphs.

Proposition 8.7. *Let X be any connected bipartite graph with biparts γ and β. There is a well defined homomorphism*

$$\sigma\colon \mathrm{Aut}(X) \to \mathrm{Sym}(\{\gamma, \beta\}),$$

taking any graph automorphism to the permutation which it induces on $\{\gamma, \beta\}$. Let $G_0 = \ker(\sigma)$, i.e.:

$$G_0 = \{\varphi \in \mathrm{Aut}(X) : \varphi(\gamma) = \gamma, \varphi(\beta) = \beta\}.$$

Then either $G_0 = \mathrm{Aut}(X)$ or G_0 has index 2 in $\mathrm{Aut}(X)$.

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Proof. Let φ be any automorphism of X. Since φ preserves distances, we see that for any vertices u, v at an even distance, the vertices $\varphi(u)$ and $\varphi(v)$ are at an even distance. Since X is connected, two vertices are at an even distance if and only if they belong to the same set in the bipartition, and thus we see that $\varphi \in G_0$ or $\varphi(\beta) = \gamma$ and $\varphi(\gamma) = \beta$. Hence, the group homomorphism $\sigma : \mathrm{Aut}(X) \to \mathrm{Sym}(\{\gamma, \beta\})$ is well defined, and since $\ker(\sigma) = G_0$, the result follows. □

Theorem 8.8. *Let* $\mathrm{Cin}(G, \pi) = (\gamma \cup \beta, E)$ *be a connected* β*-regular Cayley incidence graph with* $\pi = \{C_1, C_2, \ldots, C_k\}$, $C_1 = \{e, g_2, \ldots, g_k\}$, $g_1 = e$ *and* $C_i = g_i^{-1}C_1$. *Let* $\pi^* = \{C_1^*, \ldots, C_k^*\}$, *where* $C_i^* = g_i C_1^{-1}$. *If there exists a group automorphism* $\varphi : G \to G$ *such that* $\varphi(C_1) = C_j^*$ *for some* j, *then* $f_\varphi : \gamma \cup \beta \to \gamma \cup \beta$ *defined by*

$$\begin{cases} g \mapsto \varphi(g)C_1 & \text{for } g \in \gamma \\ gC_1 \mapsto \varphi(g)g_j & \text{for } gC_1 \in \beta, \end{cases}$$

is a graph automorphism, and one has

$$N_A(G) \cong \langle N_{AH}(G), f_\varphi \rangle.$$

If there does not exist such a group automorphism, then

$$N_A(G) \cong N_{AH}(G).$$

Proof. Throughout this proof, we will use the notation $\tilde{L}_g : \beta \to \beta : hC \mapsto ghC$ for the left action on β, and $\mathcal{L}_g : G \to G : h \mapsto gh$ for the left action on G. We first show that for any group automorphism $\varphi \colon G \to G$ with $\varphi(C_1) = C_j^*$, f_φ is a graph automorphism and, moreover, $f_\varphi \in N_A(G) \setminus N_{AH}(G)$. Using $\varphi(C_1) = C_j^* = g_j C_1^*$, we see that $h \in gC_1$ is equivalent to $\varphi(g^{-1}h) \in \varphi(C_1) = g_j C_1^*$, which is in turn equivalent to $f_\varphi(gC_1) = \varphi(g)g_j \in \varphi(h)C_1 = f_\varphi(h)$, so f_φ is a graph homomorphism. Since f_φ is a bijection, f_φ is a graph automorphism. Further, we have

$$(f_\varphi)_{|\gamma} \circ \mathcal{L}_g = \tilde{L}_{\varphi(g)} \circ (f_\varphi)_{|\gamma},$$
$$(f_\varphi)_{|\beta} \circ \tilde{L}_g = \mathcal{L}_{\varphi(g)} \circ (f_\varphi)_{|\beta},$$

so $f_\varphi \in N_A(G)$. Finally, since f_φ swaps the biparts, we have $f_\varphi \in N_A(G) \setminus N_{AH}(G)$.

Consider the map $\sigma \colon A \to \mathrm{Sym}(\{\gamma, \beta\})$ defined in Proposition 8.7. Since $N_A(G) \cap \ker(\sigma) = N_{AH}(G)$, by Proposition 8.7 we see that either $N_A(G)$ is an extension of $N_{AH}(G)$ by $\mathbb{Z}_2$ or $N_{AH}(G) = N_A(G)$. We are in the first case if there exists any element $f \in N_A(G) \setminus N_{AH}(G)$, and in the second case otherwise. To prove the theorem, it remains to show that if there is a graph automorphism $f \in N_A(G)$ that swaps the biparts, then there exists a group automorphism $\varphi \colon G \to G$ with $\varphi(C_1) = C_j^*$ for some j.

Let f be a graph automorphism $f \in N_A(G) \setminus N_{AH}(G)$. After applying, if necessary, a translation by an element of G, we may assume that $f(e) = C_1$. Since $\mathrm{Cin}(G, \pi)$ is β-regular, the map $\tau : \beta \to G : gC_1 \mapsto g$ is a bijection, and $\tau \circ \tilde{L}_g \circ \tau^{-1} = \mathcal{L}_g$. Then, $\varphi = \tau \circ f_{|\gamma}$ is a bijection $\varphi : G \to G$, and $f(e) = C_1$ implies that $\varphi(e) = e$. Then, since $f \in N_A(G)$, we see that $f_{|\gamma} \circ \mathcal{L}_g \circ (f_{|\gamma})^{-1} = \tilde{L}_h$, hence

$$\varphi \circ \mathcal{L}_g \circ \varphi^{-1} = \tau \circ f_{|\gamma} \circ \mathcal{L}_g \circ (f_{|\gamma})^{-1} \circ \tau^{-1} = \tau \circ \tilde{L}_h \circ \tau^{-1} = \mathcal{L}_h,$$

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and thus, by Lemma 8.1, φ is a group automorphism.

Since f maps edges to edges, we see that $f(C_1)$ is adjacent to $f(e) = C_1$, i.e. $f(C_1) = g_j$ for some j. The neighbourhood of C_1, i.e. $\{e, g_2, \ldots, g_k\} \subseteq \gamma$, is mapped under f to the neighbourhood of g_j, that is, the set $\{g_j g_i^{-1} C_1 : i \in \{1, \ldots, k\}\}$, thus

$$\begin{aligned}\varphi(\{e, g_2, \ldots, g_k\}) &= \tau(\{g_j g_i^{-1} C_1 : i \in \{1, \ldots, k\}\}) \\ &= \{g_j g_i^{-1} : i \in \{1, \ldots, k\}\},\end{aligned}$$

and hence $\varphi(C_1) = C_j^*$, as required. □

Example 8.9. We consider the quaternion group Q_8, with

$$\pi = \{\{1, i, -j\}, \{1, -i, k\}, \{1, j, -k\}\},$$

and we let $\varphi : Q_8 \to Q_8$ be given by $i \mapsto j$, $j \mapsto i$, $k \mapsto -k$. Then π' is given by

$$\pi' = \{\{1, -i, j\}, \{1, i, k\}, \{1, -j, k\}\}$$

This induces an automorphism of $\mathrm{Cin}(Q_8, \pi)$ which swaps the two sides.

9 Bipartite Cayley graphs and bi-Cayley graphs

We constructed the Fano Plane in Example 2.7 as a Cayley incidence graph, but we can also construct it as a regular Cayley graph. Consider the dihedral group of order 14

$$D_7 = \langle a, b : a^n = b^2 = e, bab = a^{-1} \rangle,$$

and let $T = \{b, ab, a^3 b\}$. Then $\mathrm{Cay}(D_7, T)$ is isomorphic to

$$\mathrm{Cin}(\mathbb{Z}_7, \{\{0, 1, 3\}, \{0, 4, 5\}, \{0, 2, 6\}\}).$$

More generally, it is natural to ask when a regular Cayley incidence graph is a bipartite Cayley graph, or vice-versa.

We will need the following fact about bipartite Cayley graphs.

Proposition 9.1. *A Cayley graph $\mathrm{Cay}(G, S)$ is bipartite if and only if there exists a homomorphism $\varphi\colon G \to \mathbb{Z}_2$ such that $\varphi(S) \subseteq \{1\}$.*

Proof. Given such a homomorphism, it is clear that $\varphi^{-1}(0)$ and $\varphi^{-1}(1)$ define a bipartition of $V(\mathrm{Cay}(G, S))$.

Conversely, suppose that we are given a Cayley graph which is bipartite. We consider the homomorphism $\varphi\colon G \to \mathbb{Z}_2$ which is the homomorphism σ as in Proposition 8.7, restricted to G. It is surjective since G acts transitively on $\mathrm{Cay}(G, S)$. By the way that elements of S act on $\mathrm{Cay}(G, S)$, we also see that all elements of S must swap the sides, and hence we have $\varphi(S) \subseteq \{1\}$. □

We begin by establishing that any bipartite Cayley graph with girth at least six is a Cayley incidence graph.

Theorem 9.2. *If $X = \mathrm{Cay}(G, S_1)$ is bipartite with girth at least six, then $X \cong \mathrm{Cin}(G_0, \pi)$ for some subgroup $G_0 \leq G$ of index two and some π with $S(\pi) = S_1$.*

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Proof. Since X has no four-cycles, we know that for all $s_i \neq s_j \in S_1$ the products $s_i s_j$ must be distinct.

Let $S_2 = \{s_i s_j \in G : 1 \leq i, j \leq \ell\}$, and define $\pi = \{C_1, \ldots, C_\ell\}$ by setting $C_i = \{s_i s_j : 1 \leq j \leq \ell\}$. So

$$\begin{aligned} C_1 &= \{s_1^2, \ldots, s_1 s_\ell\}, \\ C_2 &= \{s_2 s_1, s_2^2, \ldots, s_2 s_\ell\}, \\ &\vdots \\ C_\ell &= \{s_\ell s_1, \ldots, s_\ell^2\}. \end{aligned}$$

Note that each non-identity element in S_2 appears exactly once, and the identity is in every cell because S_1 is inverse-closed.

Let $1 \leq i \leq \ell$, and let $t \in C_i$. Since S_1 is inverse-closed, we can write $t = s_i s_j^{-1}$ for some $1 \leq j \leq \ell$. Then

$$t^{-1} C_i = \left\{t^{-1} s_i s_1, t^{-1} s_i s_2, \ldots, t^{-1} s_i s_\ell\right\} = \{s_j s_1, s_j s_2, \ldots, s_j s_\ell\} = C_j \in \pi,$$

so π satisfies the T-axiom.

Since $\mathrm{Cay}(G, S_1)$ is bipartite, by Proposition 9.1, we have a homomorphism $\varphi : G \to \mathbb{Z}_2$ such that $S_1 \subseteq \varphi^{-1}(1)$. We let $G_0 = \ker(\varphi)$. It follows from $S_1 \subseteq \varphi^{-1}(1)$ that $S_2 \subseteq \ker(\varphi) = G_0$.

We now show that $\mathrm{Cin}(G_0, \pi) = (\gamma \cup \beta, E)$ is isomorphic to $\mathrm{Cay}(G, S_1)$. We define $\psi \colon \gamma \cup \beta \to G$ by

$$\begin{cases} h \mapsto h & h \in \gamma. \\ hC_i \mapsto hs_i & hC_i \in \beta. \end{cases}$$

We first show that this map is well defined. This needs to be verified for vertices $x \in \beta$. Suppose that $h_1 C_i = h_2 C_j$. Then $h_1^{-1} h_2 \in C_i$ and $h_2^{-1} h_1 \in C_j$, but this is only possible if $h_1^{-1} h_2 = s_i s_j^{-1}$. Then $h_1 s_i = h_2 s_j$, as desired.

Next, we show it is bijective. Suppose that $\psi(gC_i) = \psi(hC_j)$. Then, $gs_i = hs_j$ and thus $g^{-1}h = s_i s_j^{-1} \in C_i$ and $h^{-1}g = s_j s_i^{-1} \in C_j$. But this implies that $gC_i = hC_j$. This shows that ψ is injective, and since both sets have the same cardinality, we have a bijection.

It is clear that the image of any edge in $\mathrm{Cin}(G_0, \pi)$ is an edge in $\mathrm{Cay}(G, S_1)$, and that any edge in $\mathrm{Cay}(G, S_1)$ is the image of some edge in $\mathrm{Cin}(G_0, \pi)$, and hence ψ is a graph isomorphism. □

Remark 9.3. We could also prove Theorem 9.2 using Corollary 4.5.

We now consider when a Cayley incidence graph is a Cayley graph. It clearly needs to be regular, but it is not the case that a regular Cayley incidence graph is always a Cayley graph, as evidenced by Example 6.12. Below we give some sufficient conditions for when this is the case.

Theorem 9.4. *Let* $\mathrm{Cin}(G, \pi)$ *be a connected* β*-regular Cayley incidence graph with* $\pi = \{C_1, C_2, \ldots, C_k\}$, $C_1 = \{e, g_2, \ldots, g_k\}$, $g_1 = e$ *and* $C_i = g_i^{-1} C_1$. *Let* $C_i^* = g_i C_1^{-1}$ *for* $1 \leq i \leq k$. *If there exists a group automorphism* $\varphi \colon G \to G$ *of order* 2 *such that* $\varphi(C_1) = C_j^*$, *then* $\mathrm{Cin}(G, \pi)$ *is a Cayley graph for the group* $G \rtimes \langle \varphi \rangle$.

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Proof. The group automorphism φ defines a graph isomorphism by Theorem 8.8. It is easy to see that $G \rtimes \langle\varphi\rangle$ acts regularly on the vertices of $\mathrm{Cin}(G, \pi)$. Then by Sabidussi's theorem (Theorem 4.1), we see that $\mathrm{Cin}(G, \pi)$ is a Cayley graph. □

For any abelian group G, the *generalized dihedral group* $\mathrm{Dih}(G)$ is the semidirect product $G \rtimes \mathbb{Z}_2$ with $ag = g^{-1}a$ for the non-identity element $a \in \mathbb{Z}_2$ and $g \in G$.

Corollary 9.5. *Let $X = \mathrm{Cin}(G, \pi)$ be a connected Cayley incidence graph for an abelian group G. If X is β-regular, then X is a Cayley graph of $\mathrm{Dih}(G)$.*

Proof. Unless G has exponent two, there is a group automorphism of G of order two that maps g to g^{-1} so the result follows by Theorem 9.4. If $G = \mathbb{Z}_2^d$ for some d, it follows from Corollary 8.6 that the two halved graphs of X are isomorphic. Letting φ be a graph isomorphism between them, it is easily verified that

$$G \times \langle\varphi\rangle \cong \mathbb{Z}_2^{d+1} \cong \mathrm{Dih}(G)$$

acts regularly on X. □

Corollary 9.6. *A regular Cayley incidence graph of the cyclic group $\mathbb{Z}_n$ is a Cayley graph for the dihedral group of order $2n$.*

Proof. By Lemma 6.10, a regular $\mathrm{Cin}(G, \pi)$ with abelian G is either β-transitive, or π is a collection of k subgroups of size k. The latter is impossible for $G = \mathbb{Z}_n$, so Corollary 9.5 applies. □

From Theorem 9.2 and Theorem 9.4, we see that Cayley incidence graphs generalize bipartite Cayley graphs of girth at least six. We can extend this generalization to the related concept of bi-Cayley graphs.

Definition 9.7. A *bi-Cayley graph*, sometimes called a *semi-Cayley graph*, is a graph which admits a semiregular group action such that there are exactly two vertex orbits of equal size.

Bi-Cayley graphs are a natural way of loosening Sabidussi's theorem (Theorem 4.1) to include large classes of highly symmetric graphs. There has been much work on bi-Cayley graphs, from early results characterizing when bi-Cayley graphs were strongly regular [12, 43, 46] to more recent work on the spectrum [3, 28], particular classes of bi-Cayley graphs [11, 59] and their properties [17, 57]. The work on bi-Cayley graphs is much more extensive than the examples listed here, but in the remainder of this section we are primarily concerned with comparing bi-Cayley graphs and their automorphisms to Cayley incidence graphs.

A bi-Cayley graph $\mathrm{BiCay}(G, R, L, S)$ is the graph with vertex set $G \times \{0, 1\}$, and edge set

$$\begin{aligned}&\{(g, 0) \sim (gr, 0) \ : \ g \in G, r \in R\}\\ \cup\ &\{(g, 1) \sim (gl, 1) \ : \ g \in G, l \in L\}\\ \cup\ &\{(g, 0) \sim (gs, 1) \ : \ g \in G, s \in S\}.\end{aligned}$$

Thus any bipartite bi-Cayley graph such that the two orbits are the two biparts has the form $\mathrm{BiCay}(G, \emptyset, \emptyset, S)$. Bi-Cayley graphs of the form $\mathrm{BiCay}(G, \emptyset, \emptyset, S)$ are also known as Haar graphs (introduced for abelian groups in [34]).

It is easy to see that a β-regular Cayley incidence graph is a bi-Cayley graph. The converse also holds when the bi-Cayley graph is bipartite with girth at least six.

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Theorem 9.8. *Let $X = \mathrm{BiCay}(G, \emptyset, \emptyset, S)$, where $S = \{s_1, \ldots, s_\ell\}$. If X has girth at least six, then X is isomorphic to $\mathrm{Cin}(G, \pi)$, where $\pi = \{C_1, \ldots, C_\ell\}$, with*

$$\begin{aligned} C_1 &= \{e, s_1^{-1}s_2, \ldots, s_1^{-1}s_\ell\}, \\ C_2 &= \{s_2^{-1}s_1, e, \ldots, s_2^{-1}s_\ell\}, \\ &\vdots \\ C_\ell &= \{s_\ell^{-1}s_1, s_\ell^{-1}s_2, \ldots, e\}. \end{aligned}$$

Proof. It is easy to verify that π satisfies the T-axiom. We define a graph isomorphism from $\mathrm{BiCay}(G, \emptyset, \emptyset, S)$ to $\mathrm{Cin}(G, \pi)$ by

$$\begin{cases} (g, 0) \mapsto g, \\ (h, 1) \mapsto (hs_1)C_1. \end{cases}$$

We now verify that this is an isomorphism. We see that $\{(g, 0), (h, 1)\}$ is an edge in $\mathrm{BiCay}(G, \emptyset, \emptyset, S)$ if and only $h = gs$, where $s \in S$. On the other hand, we see that if g and $(hs_1)C_1$ share an edge in $\mathrm{Cin}(G, \pi)$ if and only if $g = (hs_1)(s_1^{-1}s_j)$, for some j, i.e. if $g = hs_j$. □

It can be shown that Theorem 8.8 is essentially equivalent to Theorem 1.1 from Zhou and Feng [60] applied to bipartite bi-Cayley graphs, though we omit the details.

10 Open problems

There is a well-developed theory of Cayley graphs, and it would be nice to develop a parallel theory for Cayley incidence graphs. We conclude by highlighting some open problems inspired by the work in the paper and the tables in the appendices.

Every group admits a Cayley graph, but the T-axiom is more restrictive, and there are many groups that do not admit a non-trivial collection π satisfying the conditions of Definition 2.3, as summarized in Table 1.

Question 10.1. When is a Cayley graph the underlying Cayley graph of a non-trivial Cayley incidence graph?

Let $X = \mathrm{Cay}(G, S)$. For X to be the underlying Cayley graph of a nontrivial Cayley incidence graph, we have to be able to partition the connection set into ℓ cells of size $k - 1$, for $\ell \geq 2$ and $k \geq 3$. Further, X must contain cliques of size k and, by Proposition 3.3, the minimal eigenvalue of X must be at least $-\ell$. This imposes some necessary conditions, but as is clear from Table 1, the structure of the group plays a vital role in determining how many π satisfy the T-axiom.

If a full characterization is out of reach, it would be helpful to be to build up Cayley incidence graphs as we did in Lemma 3.5 and Lemma 3.7.

Question 10.2. What operations on groups and cells preserve the T-axiom?

In the small examples listed in Table 4, every regular Cayley incidence graph is isomorphic to a Cayley graph, but Example 6.12 shows this need not be the case. Thus we might ask to build on Table 4 and the work in Section 9 to answer the following question.

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Question 10.3. When is a regular Cayley incidence graph not a Cayley graph?

There has been some work on such questions, under the name of Haar graphs [20, 24, 25].

It would also be interesting to further develop the connections between Cayley incidence graphs and the work in related areas, including hypergraphs, design theory, and bi-Cayley graphs.

Cayley hypergraphs have been introduced a number of times [9, 38, 40, 52], and we can ask how the new class of Cayley incidence graphs can be used in relation to the problems considered by those authors. Of particular note, we highlight the biregular cages introduced in Section 7.

Problem 10.4. Use Cayley incidence graphs to find biregular cages.

Section 5 discussed the use of difference sets and partial difference families to construct projective and affine spaces as Cayley incidence graphs.

Question 10.5. What other structures from design theory and finite geometry can be constructed using Cayley incidence graphs?

A key property of Cayley graphs is that they give many examples of vertex-transitive graphs. This suggests that the β-transitive graphs, introduced in Section 6, would provide an especially good extension of Cayley graphs to a bipartite context.

Problem 10.6. Further develop the theory of β-transitive Cayley incidence graphs.

Finally, there is a beautiful theory of graphical regular representations of Cayley graphs, and it would be interesting to try and extend it to Cayley incidence graphs.

Problem 10.7. When is the automorphism group of a Cayley incidence graph $\mathrm{Cin}(G, \pi)$ isomorphic to G?

For Haar graphs, which is the β-regular case, this question has been studied [27, 48], though the general case has not. We could also reframe this problem in terms of hypergraphs.

Problem 10.8. When is the automorphism group of the hypergraph with incidence graph $\mathrm{Cin}(G, \pi)$ isomorphic to G?

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Appendix

Order	Group	GAP index	Non-trivial π
7	$\mathbb{Z}_7$	$(7,1)$	1
8	$\mathbb{Z}_8$	$(8,1)$	1
8	$\mathbb{Z}_4 \times \mathbb{Z}_2$	$(8,2)$	0
8	$\mathbb{Z}_2^3$	$(8,5)$	0
8	D_4	$(8,3)$	0
8	Q_8	$(8,4)$	1
9	$\mathbb{Z}_9$	$(9,1)$	1
9	$\mathbb{Z}_3^2$	$(9,2)$	3
10	$\mathbb{Z}_{10}$	$(10,2)$	1
10	D_5	$(10,1)$	0
11	$\mathbb{Z}_{11}$	$(11,1)$	1
12	$\mathbb{Z}_{12}$	$(12,2)$	4
12	$\mathbb{Z}_6 \times \mathbb{Z}_2$	$(12,5)$	2
12	D_6	$(12,4)$	0
12	Dic_3	$(12,1)$	3
12	A_4	$(12,3)$	4
13	$\mathbb{Z}_{13}$	$(13,1)$	4
14	$\mathbb{Z}_{14}$	$(14,2)$	3
14	D_7	$(14,1)$	0
15	$\mathbb{Z}_{15}$	$(15,1)$	16
16	$\mathbb{Z}_{16}$	$(16,1)$	8
16	$\mathbb{Z}_4^2$	$(16,2)$	6
16	$\mathbb{Z}_8 \times \mathbb{Z}_2$	$(16,5)$	1
16	$\mathbb{Z}_4 \times \mathbb{Z}_2^2$	$(16,10)$	1
16	$\mathbb{Z}_2^4$	$(16,14)$	4
16	D_8	$(16,7)$	0
16	Dic_4	$(16,9)$	5
16	$\mathbb{Z}_4 \rtimes \mathbb{Z}_4$	$(16,4)$	3
16	$(\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_4$	$(16,3)$	4
16	$\mathbb{Z}_8 \rtimes \mathbb{Z}_2$	$(16,6)$	5
16	QD_8	$(16,8)$	2
16	$D_4 \times \mathbb{Z}_2$	$(16,11)$	2
16	$Q_8 \times \mathbb{Z}_2$	$(16,12)$	2
16	$(\mathbb{Z}_4 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_2$	$(16,13)$	0

Table 1: List of all groups of order at most 16 (and at least seven) and the number of non-isomorphic, non-trivial Cayley incidence graphs for each. Here, D_n is a dihedral group, Dic_n a dicyclic group, Q_8 the quaternion group, A_4 the alternating group, QD_8 the quasi-dihedral group, $Q_8 \times \mathbb{Z}_2$ the Hamiltonian group, and $(\mathbb{Z}_4 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_2$ the Pauli group. The third column further gives the group ID in GAP's small group library.

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Group	ℓ	k	Cayley	Order of Aut	Orbits of Aut
$\mathbb{Z}_8$	3	3	Yes	96	1
$\mathbb{Z}_9$	3	3	Yes	18	1
$\mathbb{Z}_{10}$	3	3	Yes	20	1
$\mathbb{Z}_{11}$	3	3	Yes	22	1
$\mathbb{Z}_{12}$	3	3	Yes	24	1
$\mathbb{Z}_{12}$	3	3	Yes	24	1
$\mathbb{Z}_{12}$	3	3	Yes	48	1
$\mathbb{Z}_{12}$	4	3	No	12	3
$\mathbb{Z}_{13}$	3	3	Yes	26	1
$\mathbb{Z}_{13}$	3	3	Yes	78	1
$\mathbb{Z}_{13}$	6	3	No	39	3
$\mathbb{Z}_{13}$	4	4	Yes	11232	1
$\mathbb{Z}_{14}$	3	3	Yes	28	1
$\mathbb{Z}_{14}$	3	3	Yes	28	1
$\mathbb{Z}_{14}$	4	4	Yes	672	1
$\mathbb{Z}_{15}$	3	3	Yes	30	1
$\mathbb{Z}_{15}$	3	3	Yes	30	1
$\mathbb{Z}_{15}$	3	3	Yes	60	1
$\mathbb{Z}_{15}$	3	3	Yes	60	1
$\mathbb{Z}_{15}$	4	3	No	15	3
$\mathbb{Z}_{15}$	4	3	No	30	3
$\mathbb{Z}_{15}$	6	3	No	15	3
$\mathbb{Z}_{15}$	6	3	No	15	3
$\mathbb{Z}_{15}$	6	3	No	15	3
$\mathbb{Z}_{15}$	6	3	No	360	2
$\mathbb{Z}_{15}$	6	3	No	60	2
$\mathbb{Z}_{15}$	4	4	Yes	30	1
$\mathbb{Z}_{15}$	4	4	Yes	60	1
$\mathbb{Z}_{15}$	4	4	Yes	720	1
$\mathbb{Z}_{15}$	7	3	No	60	3
$\mathbb{Z}_{15}$	7	3	No	$20160 = 8!/2$	2

Table 2: All non-trivial Cayley incidence graphs (up to graph isomorphism) for cyclic groups of order at most 15.

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Group	ℓ	k	Cayley	Order of Aut	Orbits of Aut
$\mathbb{Z}_3^2$	2	3	No	72	2
$\mathbb{Z}_3^2$	3	3	Yes	216	1
$\mathbb{Z}_3^2$	4	3	No	432	2
$\mathbb{Z}_6 \times \mathbb{Z}_2$	3	3	Yes	144	1
$\mathbb{Z}_6 \times \mathbb{Z}_2$	4	3	No	576	2
$\mathbb{Z}_4^2$	3	3	Yes	192	1
$\mathbb{Z}_4^2$	2	4	No	1152	2
$\mathbb{Z}_4^2$	3	4	No	192	2
$\mathbb{Z}_4^2$	6	3	No	96	2
$\mathbb{Z}_4^2$	4	4	Yes	2304	1
$\mathbb{Z}_4^2$	5	4	No	5760	2
$\mathbb{Z}_8 \times \mathbb{Z}_2$	3	3	Yes	64	1
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	2	4	No	1152	2
$\mathbb{Z}_2^4$	2	4	No	1152	2
$\mathbb{Z}_2^4$	3	4	No	576	2
$\mathbb{Z}_2^4$	4	4	Yes	2304	1
$\mathbb{Z}_2^4$	5	4	No	5760	2

Table 3: All non-trivial Cayley incidence graphs (up to graph isomorphism) for abelian, non-cyclic groups of order at most 16.

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Group	ℓ	k	Girth	Cayley	Order of Aut	Orbits of Aut
Q_8	3	3	6	Yes	96	1
Dic_{12}	3	3	6	Yes	48	1
Dic_{12}	3	3	6	Yes	144	1
Dic_{12}	4	3	6	No	576	2
A_4	2	3	8	No	48	2
A_4	3	3	6	Yes	144	1
A_4	2	4	6	No	48	2
A_4	4	3	6	No	576	2
Dic_4	3	3	6	Yes	64	1
Dic_4	6	3	6	No	32	3
Dic_4	4	4	6	Yes	2304	1
Dic_4	4	4	6	Yes	64	1
Dic_4	5	4	6	No	5760	2
$\mathbb{Z}_4 \rtimes \mathbb{Z}_4$	3	3	6	Yes	64	1
$\mathbb{Z}_4 \rtimes \mathbb{Z}_4$	2	4	8	No	1152	2
$\mathbb{Z}_4 \rtimes \mathbb{Z}_4$	6	3	6	No	96	2
$\mathbb{Z}_2^2 \rtimes \mathbb{Z}_4$	2	4	8	No	1152	2
$\mathbb{Z}_2^2 \rtimes \mathbb{Z}_4$	3	4	6	No	576	2
$\mathbb{Z}_2^2 \rtimes \mathbb{Z}_4$	4	4	6	Yes	2304	1
$\mathbb{Z}_2^2 \rtimes \mathbb{Z}_4$	5	4	6	No	5760	2
$\mathbb{Z}_8 \rtimes \mathbb{Z}_2$	2	4	8	No	1152	2
$\mathbb{Z}_8 \rtimes \mathbb{Z}_2$	3	4	6	No	192	2
$\mathbb{Z}_8 \rtimes \mathbb{Z}_2$	3	3	6	Yes	192	1
$\mathbb{Z}_8 \rtimes \mathbb{Z}_2$	4	4	6	Yes	2304	1
$\mathbb{Z}_8 \rtimes \mathbb{Z}_2$	5	4	6	No	5760	2
QD_8	2	4	8	No	1152	2
QD_8	3	4	6	No	192	2
$D_4 \times \mathbb{Z}_2$	2	4	8	No	1152	2
$D_4 \times \mathbb{Z}_2$	3	4	6	No	192	2
$Q_8 \times \mathbb{Z}_2$	4	4	6	Yes	2304	1
$Q_8 \times \mathbb{Z}_2$	5	4	6	No	5760	2

Table 4: All non-trivial Cayley incidence graphs (up to graph isomorphism) for non-abelian groups of order at most 16 with D_n a dihedral group, Dic_n a dicyclic group, Q_8 the quaternion group, A_4 the alternating group, QD_8 the quasi-dihedral group, $Q_8 \times \mathbb{Z}_2$ the Hamiltonian group, and $(\mathbb{Z}_4 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_2$ the Pauli group.

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